

# The energy-time uncertainty principle and quantum phenomena

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We employ the energy-time uncertainty principle to provide heuristic yet helpful insights into tunneling, Unruh radiation, the Schwinger effect, and the ground state of the electromagnetic field. The position-momentum uncertainty principle is employed in auxiliary roles. We also discuss the similarities and differences between quantum and thermal fluctuations. © 2010 American Association of Physics Teachers.

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## I. INTRODUCTION

The energy-time uncertainty principle is not considered to be as well established as the position-momentum uncertainty principle. There are many different interpretations of the energy-time uncertainty principle,<sup>1-5</sup> and it has been derived from first principles by Kobe and Aguilera-Navarro.<sup>3,4</sup>

The position-momentum uncertainty principle can be most easily understood by considering a wave train that is truncated into a packet localized within a finite spatial interval  $L = \Delta x$ . This packet cannot with certainty have a wavelength  $\lambda$  equal to its expectation value  $\langle \lambda \rangle$ , and hence momentum  $p = h/\lambda$  equal to its expectation value  $\langle p \rangle = h/\langle \lambda \rangle$ . There must be relative uncertainties  $\Delta \lambda / \langle \lambda \rangle \approx \langle \lambda \rangle / L$  and hence  $\Delta p / \langle p \rangle \approx (h/L) / (h/\langle \lambda \rangle) = \langle \lambda \rangle / L \approx \Delta \lambda / \langle \lambda \rangle$ , which implies that  $\Delta p \Delta x = \Delta p L \approx \langle p \rangle (\langle \lambda \rangle / L) L = \langle p \rangle \langle \lambda \rangle = (h/\langle \lambda \rangle) \langle \lambda \rangle = h$ . Similarly, the energy-time uncertainty principle can be understood as arising from the truncation of a wave train into a packet localized within a finite temporal interval  $\Delta t$ . This packet cannot with certainty have frequency  $\nu$  equal to its expectation value  $\langle \nu \rangle$ , and hence energy  $E = h\nu$ . We must have  $\Delta \nu / \langle \nu \rangle = \Delta E / \langle E \rangle \approx (1/\Delta t) / \langle \nu \rangle = 1/(\langle \nu \rangle \Delta t)$ , which implies that  $\Delta E \Delta t \approx \langle E \rangle [1/(\langle \nu \rangle \Delta t)] \Delta t = \langle E \rangle (1/\langle \nu \rangle) = h \langle \nu \rangle \times (1/\langle \nu \rangle) = h$ .

We accept the relations  $p = h/\lambda$  and  $E = h\nu$  as axioms. We do not know why a wave train or packet must have momentum and energy.<sup>6</sup> Given that a wave train or packet has momentum and energy, we do not fully understand why momentum is inversely related to wavelength and why energy is directly related to frequency.<sup>6</sup> However we have partial understanding of the proportionalities  $p \propto 1/\lambda$  and  $E \propto \nu$  because they are suggested by relativity.<sup>6</sup>

Recent papers on tunneling<sup>7</sup> and on Unruh radiation<sup>8</sup> have inspired this paper. Several applications of the energy-time uncertainty principle to nonrelativistic, relativistic, and (hypothetical) tachyonic tunneling<sup>7</sup> are discussed in Sec. II. Applications to Unruh radiation<sup>8</sup> and the Schwinger effect<sup>9</sup> are discussed in Sec. III. Work of Cohen<sup>10</sup> on tunneling pioneered the consideration not only of the potential energy required to surmount a barrier during tunneling but of the kinetic energy required as well. We will extend the method of Cohen<sup>10</sup> in our discussions of tunneling, Unruh radiation, and the Schwinger effect. The position-momentum uncertainty principle is employed in auxiliary roles. We obtain approximate expressions for the tunneling speed and related quantities, the temperature of Unruh radiation, and the electric field strength required for the Schwinger effect. Some related phenomena are also discussed. The ground state of

the electromagnetic field, emission, and absorption are qualitatively discussed in Sec. IV. Quantum and thermal fluctuations and their respective probability distributions are compared and contrasted in Sec. V. Possible pedagogical applications of our results are discussed in Sec. VI. Supplementary points are discussed in Appendices A and B.

## II. THE ENERGY-TIME UNCERTAINTY PRINCIPLE AND TUNNELING

### A. Nonrelativistic tunneling

Davies<sup>7</sup> applied the energy-time uncertainty principle<sup>1-3</sup> as follows. Let  $\Delta \mathcal{E}_{\text{rms}} \equiv ((\mathcal{E} - \langle \mathcal{E} \rangle)^2)^{1/2} = ((\mathcal{E}^2) - \langle \mathcal{E} \rangle^2)^{1/2}$  be the root-mean-square (rms) measure of energy nonconservation and  $\Delta t$  be the time interval during which this nonconservation is sustained. Take  $\Delta \mathcal{E}_{\text{rms}} \Delta t$  to be on the order of the rms uncertainty-principle minimum<sup>1-3,7</sup>

$$\Delta \mathcal{E}_{\text{rms}} \Delta t \approx (\Delta \mathcal{E}_{\text{rms}} \Delta t)_{\text{min}} = \frac{1}{2} \hbar. \quad (1)$$

Davies<sup>7</sup> considered a heuristic argument in which a particle of mass  $m$  with speed  $v \ll c$  and kinetic energy  $E = mv^2/2$  approaches a square potential barrier of height  $V > E$  and width  $a$  and borrows energy  $\Delta \mathcal{E} = V - E$  to surmount this barrier. But if the particle borrows only  $\Delta \mathcal{E} = V - E$ , it would have zero kinetic energy and hence zero speed while traversing the barrier, and therefore would require  $\Delta t = \infty$  to traverse the barrier. Hence tunneling would be impossible.<sup>10</sup> Therefore, the particle's total borrowed energy should be

$$\Delta \mathcal{E} = V - E + K = V - E + mu^2/2, \quad (2)$$

where  $u$  is its speed and  $K = mu^2/2$  is its borrowed kinetic energy while traversing the barrier. Then the barrier is traversed in time  $\Delta t = a/u$ , and hence

$$\Delta \mathcal{E} \Delta t = (V - E + mu^2/2) \Delta t = (V - E + mu^2/2) \frac{a}{u}. \quad (3)$$

The probability  $P(\geq \Delta \mathcal{E} \Delta t)$  that energy of magnitude  $\Delta \mathcal{E}$  or more can be borrowed (or discarded) via quantum-mechanical vacuum fluctuations associated with Eq. (1) (Refs. 11–15) for time  $\Delta t$  or longer decreases quasi-exponentially:<sup>10</sup>

$$P(\geq \Delta \mathcal{E} \Delta t) \sim e^{-\Delta \mathcal{E} \Delta t / (\Delta \mathcal{E}_{\text{rms}} \Delta t)_{\text{min}}} = e^{-2 \Delta \mathcal{E} \Delta t / \hbar}. \quad (4)$$

The  $e$ -folding scale, corresponding to a decrease in  $P(\geq \Delta \mathcal{E} \Delta t)$  by a factor of  $e$ , is the rms uncertainty-principle minimum given by Eq. (1). The quasi-exponential decrease in  $P(\geq \Delta \mathcal{E} \Delta t)$  becomes exponential for nearly opaque barriers, that is, for  $P(\geq \Delta \mathcal{E} \Delta t) \ll 1$ .<sup>16</sup>

Hence, for a given  $V-E$  and  $a$ , tunneling is easiest and most probable if  $\Delta\mathcal{E}\Delta t$  is minimized. Following Cohen,<sup>10</sup> this minimization yields the most probable tunneling speed  $u^*$  and related quantities. We have

$$\frac{\partial(\Delta\mathcal{E}\Delta t)}{\partial u} = \frac{\partial}{\partial u} \left[ \left( V - E + \frac{1}{2}mu^2 \right) \frac{a}{u} \right] = 0, \quad (5)$$

which gives

$$u^* = \left[ \frac{2(V-E)}{m} \right]^{1/2} = v \left( \frac{V}{E} - 1 \right)^{1/2}, \quad (6)$$

$$\Delta\mathcal{E}^* = V - E + K^* = V - E + \frac{1}{2}mu^{*2} = 2(V-E) = mu^{*2}, \quad (7)$$

$$\Delta t^* = \frac{a}{u^*} = a \left[ \frac{m}{2(V-E)} \right]^{1/2} = \frac{a}{v} \left( \frac{V}{E} - 1 \right)^{-1/2}, \quad (8)$$

and

$$(\Delta\mathcal{E}\Delta t)^* = (\Delta\mathcal{E}\Delta t)_{\min} = a[2m(V-E)]^{1/2} = amu^*. \quad (9)$$

The results in Eqs. (6)–(9) agree with those in Ref. 10 and with results obtained via numerous other methods.<sup>17,18</sup> These results require  $v \ll c$  and  $u \ll c$  for all  $u$  that have non-negligible probabilities of being equaled or exceeded (including  $u^* \ll c$ ).

Note that  $(\Delta\mathcal{E}\Delta t)_{\min}$  in Eq. (9) is, by Eq. (4), the most probable value of  $\Delta\mathcal{E}\Delta t$  consistent with the given values of  $V-E$  and  $a$ . This minimum value should not be confused with the minimum possible rms uncertainty-principle average  $(\Delta\mathcal{E}_{\text{rms}}\Delta t)_{\min} = \hbar/2$  in Eq. (1). For given  $V-E$ ,  $a$ , and  $u$ ,  $\Delta\mathcal{E}\Delta t$  in Eq. (3), and hence  $(\Delta\mathcal{E}\Delta t)_{\min}$  in Eq. (9), is smaller if  $u$  is constant than if  $u$  is the average value of any nonconstant  $u(t)$  (see Appendix A).

The interpretation that a particle traverses over rather than through a barrier with the help of borrowed energy emphasizes its particle aspect in contrast to its wave aspect. This interpretation is closer to being correct if the particle is well localized compared to the barrier width (that is,  $\Delta x \ll a$ ) not only while traversing the barrier but also while it is a free particle outside it and if both  $u$  and  $v$  are well-defined, that is,  $\Delta u \ll u$  and  $\Delta v \ll v$ . This condition ensures that the times when the particle enters and leaves the barrier are well-defined to within  $\ll a/v$  and  $\ll a/u$ , respectively. The position-momentum uncertainty principle requires for such localization that  $\Delta x \ll a$  and  $\Delta u \ll u$  while a particle traverses the barrier, and thus

$$\begin{aligned} p(u) &= mu \gg [\Delta p(u)]_{\text{rms},\min} \\ &= m(\Delta u)_{\text{rms},\min} = \frac{\hbar}{2\Delta x} \gg \frac{\hbar}{2a}, \end{aligned} \quad (10)$$

which implies that

$$amu \gg \hbar/2. \quad (11)$$

Similarly, for  $\Delta x \ll a$  and  $\Delta v \ll v$  to hold while the particle is outside the barrier, we must have

$$p(v) = mv \gg [\Delta p(v)]_{\text{rms},\min} = m(\Delta v)_{\text{rms},\min} = \frac{\hbar}{2\Delta x} \gg \frac{\hbar}{2a}, \quad (12)$$

which implies that

$$amv \gg \hbar/2. \quad (13)$$

If we apply the inequality  $V > E$  to Eq. (12), the position-momentum uncertainty principle yields the localization requirement

$$V > E = \frac{[p(v)]^2}{2m} \gg \frac{\hbar^2}{8ma^2}, \quad (14)$$

which implies that

$$ma^2V \gg \hbar^2/8. \quad (15)$$

We next apply the position-momentum uncertainty principle consistent with Eqs. (10) and (11), and the energy-time uncertainty principle consistent with Eqs. (4) and (9). For both  $\Delta x \ll a$  and  $\Delta u \ll u^*$  to be realized while a particle traverses the barrier, these two uncertainty principles require that

$$\begin{aligned} (\Delta\mathcal{E}\Delta t)_{\min} &= a[2m(V-E)]^{1/2} \\ &= amu^* \gg (\Delta\mathcal{E}_{\text{rms}}\Delta t)_{\min} = \hbar/2, \end{aligned} \quad (16)$$

which somewhat tightens Eq. (15) to

$$ma^2(V-E) \gg \hbar^2/8, \quad (17)$$

and, with the help of Eqs. (4) and (9), yields

$$\begin{aligned} P(\geq \Delta\mathcal{E}\Delta t)_{\min} \\ \sim e^{-2(\Delta\mathcal{E}\Delta t)_{\min}/\hbar} = e^{-[8m(V-E)]^{1/2}a/\hbar} = e^{-2amu^*/\hbar} \ll 1. \end{aligned} \quad (18)$$

Hence, the barrier must be nearly opaque to tunneling, even with  $\Delta\mathcal{E}\Delta t$  minimized as in Eqs. (5)–(9), and only a small fraction of the incident particles manages to traverse the barrier at all, and in particular with  $u \geq u^*$ .<sup>17–19</sup> Thus, the strong inequalities in Eqs. (10)–(18) are necessary for accurate experimental measurements of  $u$  (if  $u \geq u^*$ ) via time-of-flight experiments.<sup>17–19</sup> Equation (18) yields, apart from the pre-exponential factor of  $16(E/V)(1-E/V)$ , the probability of traversing nearly opaque barriers for which the strong inequalities in Eqs. (16)–(18) are satisfied.<sup>16</sup>

The interpretation that a particle traverses over rather than through a barrier with the help of borrowed energy may still be approximately correct even if the strong inequalities in Eqs. (10)–(18) are weakened to approximate equalities. If these strong inequalities are not merely weakened but reversed, then the wave aspects of tunneling dominate the particle aspects, Eqs. (4)–(9) become poor approximations, and hence different behavior may occur.<sup>19–21</sup> According to the evaluation of Hagmann<sup>18</sup> of this aspect in Ref. 10, significant departure from the quasi-exponentiality of Eq. (4) occurs if  $(\Delta\mathcal{E}\Delta t)_{\min} < (\Delta\mathcal{E}_{\text{rms}}\Delta t)_{\min} = \hbar/2$ . In this case the probability of tunneling is typically not much less than unity, so for heuristic purposes we need not require an exact expression for it. In cases for which these strong inequalities are not satisfied, tunneling can be treated via the position-momentum uncertainty principle as an alternative to the energy-time uncertainty principle.<sup>22</sup>

## B. Special relativistic tunneling

If we apply special relativity and let  $u$  be constant to minimize  $\Delta\mathcal{E}\Delta t$  (see Appendix A), we seem to obtain Eq. (4) forbidding tunneling at superluminal speeds (superluminal tunneling). Let  $m$  be a particle's mass and  $\epsilon = mc^2(1 - u^2/c^2)^{-1/2}$  be its relativistic energy. We have

$$\Delta\mathcal{E} = V - E + K = V - E + \epsilon - mc^2 \quad (19a)$$

$$= V - E + mc^2[(1 - u^2/c^2)^{-1/2} - 1], \quad (19b)$$

which implies that

$$\Delta\mathcal{E}\Delta t = \{V - E + mc^2[(1 - u^2/c^2)^{-1/2} - 1]\} \frac{a}{u}. \quad (20)$$

Minimizing  $\Delta\mathcal{E}\Delta t$  in Eq. (20) by solving  $\partial(\Delta\mathcal{E}\Delta t)/\partial u = 0$  is usually difficult. However, in any case, from Eq. (4),  $u \rightarrow c$  (for  $a > 0$  and  $m > 0$ ) implies that  $\Delta\mathcal{E}\Delta t \rightarrow \infty$  and  $P(\geq \Delta\mathcal{E}\Delta t) \sim \exp[-(\Delta\mathcal{E}\Delta t)/\hbar] \rightarrow 0$  exponentially faster than  $\Delta\mathcal{E}\Delta t \rightarrow \infty$ , which, in turn, seems to imply the impossibility of luminal, let alone superluminal, tunneling of massive particles.

The extreme relativistic limit, that is,

$$\delta u = c - u \rightarrow 0, \quad (21)$$

yields  $u = c - \delta u \rightarrow c$ ,  $\Delta t = a/u \rightarrow a/c$ ,

$$\Delta\mathcal{E} \doteq K \rightarrow mc^2(c/2\delta u)^{1/2} = \epsilon, \quad (22)$$

and

$$\Delta\mathcal{E}\Delta t \doteq K\Delta t \rightarrow mca(c/2\delta u)^{1/2} = \epsilon a/c \quad (23)$$

for finite  $V - E > 0$ . [The dot-equal sign ( $\doteq$ ) means “very nearly equal to.”]

In light of Eqs. (4) and (18), Eqs. (20) and (23) apparently allow at most luminal, but not superluminal, tunneling,<sup>23</sup> with luminal tunneling only for massless virtual particles. For the latter,  $\Delta\mathcal{E}\Delta t$  cannot be minimized with respect to  $u$  because  $u = c = \text{constant}$  (see Appendices A and B).

An interesting special case of Eq. (20) is that of a free ( $V = 0$ ) not pre-existing ( $E = 0$ ) massive virtual particle<sup>11–15</sup> that traverses at constant  $u$  the distance  $a$  before vanishing. In this case both the mass energy  $mc^2$  and kinetic energy  $K$  must be borrowed so that

$$\Delta\mathcal{E} = \epsilon = mc^2(1 - u^2/c^2)^{-1/2} \quad (24)$$

and

$$\Delta\mathcal{E}\Delta t = mc^2(1 - u^2/c^2)^{-1/2} \frac{a}{u}. \quad (25)$$

The minimization of  $\Delta\mathcal{E}\Delta t$  in Eq. (25), i.e.,

$$\frac{\partial(\Delta\mathcal{E}\Delta t)}{\partial u} = \frac{\partial}{\partial u} \left[ mc^2(1 - u^2/c^2)^{-1/2} \frac{a}{u} \right] = 0, \quad (26)$$

yields

$$u^* = c/2^{1/2}, \quad (27)$$

$$\Delta\mathcal{E}^* = \epsilon^* = mc^2 + K^* = 2^{1/2}mc^2, \quad (28)$$

$$\Delta t^* = 2^{1/2}a/c, \quad (29)$$

and

$$(\Delta\mathcal{E}\Delta t)^* = (\Delta\mathcal{E}\Delta t)_{\min} = 2amc = 2^{1/2}a\epsilon^*/c, \quad (30)$$

where  $u$  in Eqs. (25) and (26) and  $u^*$  in Eq. (27) are relative to the local inertial frame of the massive virtual particle's creation. For example, for a massive virtual exchange particle emitted by a real particle,<sup>11–15</sup>  $u$  and  $u^*$  are relative to this real particle. For a typical massive virtual particle, which traverses  $a^* \approx \hbar/mc$  before vanishing, momentum is ill-defined, that is,  $\Delta p \approx \hbar/a^* \approx mc \approx p$ .

Relativistic analogs of the strong nonrelativistic inequalities in Eqs. (10) and (11) and in Eqs. (16)–(18) without the terms containing  $V - E$  apply only for a rare massive virtual particle that manages to traverse a distance  $a \gg a^* = \hbar c/\epsilon^*$  before vanishing. In this case momentum is well-defined, that is,  $\Delta p \approx \hbar/a \ll p \approx \hbar/a^*$ , and likewise so is  $\epsilon \approx cp$ . This improbability of a well-defined momentum is consistent with virtual particles not being directly observable (although their indirect effects are observable).<sup>11–15</sup> In any case our estimate for  $\Delta\mathcal{E}$  is heuristic, probably even more so for virtual particles than for nonrelativistic tunneling as discussed in Sec. II A (see Appendix B).

From Eqs. (4) and (30) the probability that a massive virtual particle can traverse a distance  $a$  before vanishing is

$$\begin{aligned} P(a|m>0) &\sim e^{-2\Delta\mathcal{E}\Delta t/\hbar} \\ &\approx e^{-2(\Delta\mathcal{E}\Delta t)_{\min}/\hbar} \approx e^{-2(2amc)/\hbar} = e^{-4amc/\hbar}. \end{aligned} \quad (31)$$

For a massless virtual particle such as a virtual photon, consider a wave packet of average angular frequency  $\langle\omega\rangle$  and reduced wavelength  $\langle\lambda\rangle$ . In this case, we have

$$\begin{aligned} P(a|m=0) &\sim e^{-2\Delta\mathcal{E}\Delta t/\hbar} \approx e^{-[2\hbar\langle\omega\rangle(a/c)]/\hbar} \\ &= e^{-2\langle\omega\rangle a/c} \approx e^{-[2(c/\langle\lambda\rangle)a]/c} = e^{-2a/\langle\lambda\rangle}. \end{aligned} \quad (32)$$

We have assumed the exact monochromatic relation  $\omega = c/\lambda$  to hold approximately, that is, to be modified to  $\langle\omega\rangle \approx c/\langle\lambda\rangle$  for a wave packet. The exponential forms of Eqs. (31) and (32) become exact for  $P(a) \ll 1$  and are reasonably accurate for  $P(a) \lesssim 1/e$ . Even if the density of virtual particles of a given species is a significant fraction of the Planck density [one Planck mass per (Planck length)<sup>3</sup>, the total density of all virtual particles in the quantum-mechanical vacuum],  $P(a)$  in Eqs. (31) and (32) is essentially zero if the arguments of the exponents in these equations are greater than about  $10^3$ .

There is a lower limit on  $\Delta\mathcal{E}$  for  $m > 0$  but not for  $m = 0$  (because  $\langle\omega\rangle$  can be as small as we wish, at least in a spatially infinite system). Hence, a (quasi)exponential diminution of interactions mediated by massive virtual particles is superposed at distances  $a \gg \hbar/mc$  on the purely geometrical inverse-square law (assuming three-dimensional Euclidean space).<sup>24</sup> At distances  $a \ll \hbar/mc$ , only the inverse-square decrease occurs—a range that extends to infinite distance for interactions mediated by massless virtual particles. The inverse-square decrease at distances  $a \ll \hbar/mc$  can be construed as due to the thinning out of a fixed flux of mediating virtual particles over an increasing area  $4\pi a^2$ . At distances  $a \gg \hbar/mc$ , the thinning out is of a (quasi-)exponentially decaying flux.

## C. Questioning the special relativistic speed-of-light barrier

### 1. (Hypothetical) tachyonic tunneling

The strength of the energy barrier at the speed of light is the infinite height of its peak (for massive particles), as is evident by setting  $u=c$ , that is,  $\delta u=0$ , in Eqs. (22) and (24). The weakness is the infinitesimal thinness of this peak at  $\delta u=0$  in velocity space. Let us try to exploit this weakness.

Constant  $u$  (and hence also constant  $\delta u=c-u$ ) is assumed in Secs. II A and II B.  $\Delta\mathcal{E}\Delta t$  is thereby minimized for a given barrier height and width in the nonrelativistic and relativistic regimes but not in the tachyonic regime (see Appendix A).

Let a particle of (real, positive) mass  $m$ , with initial  $v_i=c-\delta v_i$  ( $\delta v_i \ll c$ ) relative to a given local inertial frame, experience in this frame via quantum fluctuations constant rectilinear acceleration  $\mathcal{A}$  from initial  $v_i=c-\delta v_i=u_i=c-\delta u_i$  to final  $u_f=c+\delta u_i$ .<sup>25,26</sup> (A quantum fluctuation need not lead to a constant acceleration. However it could because such a Feynman–Hibbs<sup>26</sup> path is possible. So, for argument's sake, let us assume this simplest of Feynman–Hibbs<sup>26</sup> paths that breach the light barrier.) In this section we take  $\delta u=c-u \geq 0$  for  $u \leq c$  and  $\delta u=u-c \geq 0$  for  $u \geq c$ , that is,  $\delta u=|u-c|$ . Also, for simplicity, we let  $\delta u_i \ll c$  and assume that there is no potential energy barrier so that  $V=0$ .

If we extend Eqs. (19a), (19b), and (20)–(23) to the tachyonic regime  $u > c$ , then  $(1-u^2/c^2)^{-1/2}$  is imaginary. If a massive particle can traverse the infinitely tall (but also infinitesimally thin) kinetic energy barrier at  $\delta u=0$  and tunnel into the tachyonic regime, then it can have a real positive total energy  $\mathcal{E}$  there if its mass  $m$  becomes imaginary (but of opposite sign). This is not unphysical because mass is not measurable in the tachyonic regime.<sup>25</sup> Thus, if we let  $m=i|m|$  at  $u > c$ , we have, upon applying Eqs. (19a), (19b), and (20)–(23),<sup>25</sup>

$$\mathcal{E}(u < c) = mc^2(1 - u^2/c^2)^{-1/2}, \quad (33)$$

$$\mathcal{E}(u > c) = i|m|c^2(1 - u^2/c^2)^{-1/2} = |m|c^2\left(\frac{u^2}{c^2} - 1\right)^{-1/2}, \quad (34)$$

and

$$\mathcal{E}(\delta u = |u - c| \ll c) \doteq |m|c^2\left(\frac{c}{2\delta u}\right)^{1/2} \doteq K. \quad (35)$$

By Eqs. (22) and (35), for  $\delta u \ll c$ ,  $\mathcal{E} \doteq K$  and  $K$  are independent of  $\text{sgn}(c-u)$  and hence symmetrical about  $u=c$  and  $\delta u=0$ . This symmetry requires that for  $u > c$ ,  $|m|$  remains unchanged, even though (because  $\{c/[2(-\delta u)]\}^{1/2}$  is imaginary)  $m$  must also be imaginary (but of opposite sign) because  $\mathcal{E} \doteq K \doteq mc^2\{c/[2(-\delta u)]\}^{1/2}$  must be real and positive.<sup>25</sup>

Thus, we obtain for  $\delta u \ll c$

$$\Delta\mathcal{E}\Delta t = \int \Delta\mathcal{E}(t)dt = \int \Delta K(t)dt = \Delta K\Delta t = \int_{c-\delta u_i}^{c+\delta u_i} \Delta K \frac{du}{\mathcal{A}} \quad (36a)$$

$$\begin{aligned} &= \int_{-\delta u_i}^0 \Delta K \frac{[-d(\delta u)]}{\mathcal{A}} + \int_0^{\delta u_i} \Delta K \frac{d(\delta u)}{\mathcal{A}} \\ &\doteq 2 \int_{-\delta u_i}^0 \Delta K \frac{d(\delta u)}{\mathcal{A}} \end{aligned} \quad (36b)$$

$$\begin{aligned} &= 2 \int_0^{\delta u_i} mc^2 \left[ \left(\frac{c}{2\delta u}\right)^{1/2} - \left(\frac{c}{2\delta u_i}\right)^{1/2} \right] \frac{d(\delta u)}{\mathcal{A}} \\ &= \frac{mc^2}{\mathcal{A}} (8c\delta u_i)^{1/2} = \frac{4\epsilon_i \delta u_i}{\mathcal{A}}, \end{aligned} \quad (36c)$$

where we applied Eq. (22) in the last step.

The quantity  $\Delta\mathcal{E}\Delta t = \Delta K\Delta t$  in Eq. (36) is finite. The infinitely tall  $\Delta K = \infty$ ,  $\Delta\mathcal{E} = \infty$  peak of the light barrier exists only at  $\delta u=0$ , with the barrier height finite everywhere on both sides of  $\delta u=0$ .<sup>25</sup> Hence, the peak at  $\delta u=0$  can be traversed in a time interval  $\delta t=0$  of zero duration. The  $(c/2\delta u)^{1/2} \sim (\delta u/c)^{-1/2}$  singularity at  $\delta u=0$  is sufficiently weak that traversal of this singular point in velocity space entails  $\Delta\mathcal{E}\Delta t = \Delta K\Delta t=0$ , as can be seen by letting  $\delta u_i \rightarrow 0$  in Eq. (36). Hence, for  $\delta u_i \neq 0$  the integrated  $\Delta\mathcal{E}\Delta t$  of Eq. (36) is finite. [Protecting the light barrier at  $u=c$  ( $\delta u=0$ ) requires a singularity in velocity space at least as strong as  $(\delta u/c)^{-1}$ . The singularity would be strong enough if, instead of constant  $\mathcal{A}$  as we assume for simplicity,  $\mathcal{A} \propto (\delta u/c)^n$  with  $n \geq 1/2$ , because then a particle of constant attempts to breach the singularity so slowly that  $\Delta\mathcal{E}\Delta t$  becomes infinite. On the other hand, if  $n < 0$ , a particle breaches it even faster than if  $\mathcal{A}$  is constant ( $n=0$ ), and hence tunneling through the light barrier would be even easier than if  $\mathcal{A}$  is constant.]

Thus, if Eqs. (4) and (33)–(36) and the associated discussions are the only consideration, then, by Eq. (4),  $\delta u=0$  can easily be breached given that  $\Delta\mathcal{E}\Delta t \leq \hbar/2$ . Requiring  $\Delta\mathcal{E}\Delta t \leq \hbar/2$  in Eq. (36) implies that

$$\mathcal{A} \geq \frac{2mc^2}{\hbar} (8c\delta u_i)^{1/2} = \frac{8\epsilon_i \delta u_i}{\hbar}, \quad (37)$$

which in turn implies that

$$\Delta t = \frac{2\delta u_i}{\mathcal{A}} \leq \frac{\hbar}{2mc^2} (\delta u_i/2c)^{1/2} = \frac{\hbar}{4\epsilon_i} \quad (38)$$

and that the effective barrier width in coordinate space (not to be confused with its infinitesimal thinness in velocity space) is

$$a_{\text{eff}} \doteq c\Delta t = c \frac{2\delta u_i}{\mathcal{A}} \leq \frac{\hbar}{2mc} (\delta u_i/2c)^{1/2} = \frac{\hbar c}{4\epsilon_i}. \quad (39)$$

Once  $\delta u=0$  has been reached, a massive particle can become a tachyon as easily as it can revert to a tardyon provided that nature does not forbid the existence of tachyons for reasons unrelated to Eqs. (4) and (33)–(39). If nature does not forbid tachyons, then perhaps our particle can tunnel into the tachyonic regime.

For an electron or positron with  $\delta v_i = \delta u_i = 1$  m/s, Eq. (37) requires for  $\Delta\mathcal{E}\Delta t \leq \hbar/2$  and hence for  $P(\geq \Delta\mathcal{E}\Delta t)$  not much less than unity,  $\mathcal{A} \geq 7 \times 10^{25}$  m/s<sup>2</sup>, corresponding, in accordance with Eqs. (38) and (39), to  $\Delta t \leq 3 \times 10^{-26}$  s and  $a_{\text{eff}} \leq 8 \times 10^{-18}$  m. (What if, instead of relying on a quantum fluctuation, an electron or positron with  $\delta v_i = 1$  m/s could be

accelerated at  $\mathcal{A} \geq 7 \times 10^{25} \text{ m/s}^2$ ?) For an astronaut of  $m = 100 \text{ kg}$  and with  $\delta v_i = \delta u_i = 10^6 \text{ m/s}$ , Eq. (37) implies that  $\mathcal{A} \geq 8 \times 10^{60} \text{ m/s}^2$ , corresponding, in accordance with Eqs. (38) and (39), to  $\Delta t \leq 3 \times 10^{-55} \text{ s}$  and to  $a_{\text{eff}} \leq 8 \times 10^{-47} \text{ m}$ . Our astronaut could easily survive because, by Eq. (37), the required  $\mathcal{A}$  would result in a displacement of  $d = \mathcal{A}(\Delta t)^2/2 = \mathcal{A}(2\delta u_i/\mathcal{A})^2/2 = 2(\delta u_i)^2/\mathcal{A} = 2(\delta v_i)^2/\mathcal{A} \approx (\hbar/mc)(\delta u_i/2c)^{3/2} = (\hbar/mc)(\delta v_i/2c)^{3/2} \leq 2 \times 10^{-49} \text{ m}$ . Because  $\mathcal{A}\Delta t = 2\delta u_i = 2v_i = 2 \times 10^6 \text{ m/s} \ll c$ , this Newtonian result for  $d$  is sufficiently accurate and is much less than the lower limit of measurability, let alone than that required to crush an astronaut. [Time intervals shorter than the Planck time, spatial displacements smaller than the Planck length, and accelerations exceeding the Planck acceleration ( $c$  per Planck time) probably should be taken with an extra grain of salt.<sup>15,27</sup>]

Breaching the light barrier in one inertial frame implies breaching it in all, because superluminal velocities Lorentz-transform into superluminal velocities (just as subluminal ones Lorentz-transform into subluminal ones). Nevertheless, we require  $\Delta \mathcal{E} \Delta t \lesssim \hbar/2$ . Since  $\delta v_i = \delta u_i$ , and hence also  $\Delta \mathcal{E} \Delta t$ , is frame-dependent, the question arises: In which frame? Perhaps, in the (unique) frame in which the cosmic background radiation is isotropic, the closest thing there is to an “absolute rest frame,” or at least to a “preferred frame.” (Almost all of the matter in the Universe, including the Earth, is in or very nearly in this frame.)

## 2. (Hypothetical) circumvention

We briefly discuss the possibility of circumvention in contrast to tunneling of the special relativistic speed-of-light barrier. Circumvention does not require exploiting the infinitesimal thinness of the light barrier as in tunneling. Possible (hypothetical) examples include circumvention of the light barrier via (a) complex speeds,<sup>28</sup> (b) conversion of tardyon-antitardyon pairs to luxon pairs and then to tachyon-antitachyon pairs, (c) creation of tachyon-antitachyon pairs at the expense of tardyon-tardyon,<sup>29</sup> tardyon-luxon, or luxon-luxon collision energy, and (d) virtual tachyon-antitachyon pair production via quantum-mechanical vacuum fluctuations, with  $\Delta \mathcal{E} \Delta t \propto (u^2/c^2 - 1)^{-1/2} (a/u)$  minimized, indeed vanishing, at  $u^* = \infty$ . (Pairs are employed to avoid violation of lepton-number or baryon-number conservation.<sup>30</sup>)

Such circumventions “sidestep” rather than traverse the entire light barrier, not merely the infinitely tall (for massive particles) but infinitesimally thin peak of this barrier at exactly  $c$ . Although such circumventions are not forbidden by their energy costs, nature probably finds other reasons for forbidding them (as it does for traversals).

Tunneling of a barrier can always be interpreted in terms of borrowed energy in light of the energy-time uncertainty principle. In contrast, for circumvention, this is often not the case, as in the examples (a)–(c) we have discussed. However sometimes it is, as in example (d). It also is unclear whether circumvention of a nonrelativistic barrier (in contrast to traversal thereof) is possible.

## 3. The light barrier, causality, and infinite energy cost

We have shown that breaching of the light barrier is not forbidden by an infinite energy cost. Tunneling of massive particles through the light barrier is not forbidden by its infinite height because infinite height is more than compen-

sated for by infinitesimal thinness. For circumvention the infinite height is not even an issue. We now show that breaching of the light barrier is also not forbidden in order to prevent violation of causality.

The signal velocity  $u_{\text{signal}} \leq c$ , with “signal” defined as “including all points of nonanalyticity, that is, new information that is not already foretold in an earlier portion of the waveform,”<sup>23,31</sup> is a stronger restriction than the minimal one that special relativity requires to ensure that causality is not violated. This minimal restriction is derived by applying the Lorentz transformation. Let  $S$  and  $S'$  be the standard inertial frames of special relativity, with  $S'$  moving in the  $+x$  direction at velocity  $0 \leq v_{\text{rel}} \leq c$  as observed in  $S$ . Thus,  $v_{\text{rel}}$  is the relative velocity between  $S$  and  $S'$ . As observed in  $S$ , let a signal propagate a distance  $\Delta x$  from the point of transmission to the point of reception in time  $\Delta t$  and hence at speed  $u_{\text{signal}} = \Delta x / \Delta t$ . As observed in  $S'$  the time required for propagation of this signal is<sup>32–35</sup>

$$\Delta t' = (1 - v_{\text{rel}}^2/c^2)^{-1/2} \left( \Delta t - \frac{v_{\text{rel}} \Delta x}{c^2} \right) \quad (40a)$$

$$= (1 - v_{\text{rel}}^2/c^2)^{-1/2} \Delta t \left( 1 - \frac{v_{\text{rel}} \Delta x}{c^2 \Delta t} \right) \quad (40b)$$

$$= (1 - v_{\text{rel}}^2/c^2)^{-1/2} \Delta t \left( 1 - \frac{v_{\text{rel}} u_{\text{signal}}}{c^2} \right). \quad (40c)$$

The minimal restriction ensuring causality is  $\text{sgn}(\Delta t') = \text{sgn}(\Delta t)$ ,<sup>32–35</sup> which implies that

$$u_{\text{signal}} \leq \frac{c^2}{v_{\text{rel}}}. \quad (41)$$

Because  $S$  could be any reference frame, this result is completely general. Note that  $u_{\text{signal}, \text{max}} = c^2/v_{\text{rel}}$  is also the de Broglie wave speed.<sup>32–35</sup>

Nature may require “causality police” that checkmate any attempt by the transmitter and/or receiver to defeat causality by increasing  $v_{\text{rel}}$  while the signal is en route.<sup>32–35</sup> Such “police” might be of the EPR-type and/or related quantum correlations. These correlations, in contrast to superluminal but not supercausal ( $c < u_{\text{signal}} \leq c^2/v_{\text{rel}}$ ) signals, are not only nonenergy and noninformation transporting but also not even observable. Hence, they can violate Lorentz invariance and be supercausal, not merely superluminal; that is, they need not be restricted even by Eq. (41), much less by  $u_{\text{signal}} \leq c$ . However such required “policing” does not seem to preclude superluminal but not supercausal informative signals that propagate at  $c < u_{\text{signal}} \leq c^2/v_{\text{rel}}$ .

Despite the fact that such superluminal but not supercausal ( $c < u_{\text{signal}} \leq c^2/v_{\text{rel}}$ ) informative signals would obey causality, no such signal has ever been observed. Perhaps it is not known how to generate or detect such signals. Much more likely, the light barrier is inviolable, but not because breaching it is forbidden to prevent violation of causality and/or by an infinite energy cost. The reasons for its almost (but perhaps not absolutely) certain inviolability seem to be deeper and more subtle than that.

## 4. The light barrier and EPR-type correlations

To satisfy the conservation laws EPR-type and related quantum correlations must be transmitted instantaneously,

not merely superluminally ( $>c$ ) or even supercausally<sup>32–35</sup> ( $>c^2/v_{\text{rel}}$ ), in all inertial frames.<sup>36</sup> This violation of Lorentz invariance does not contradict special relativity, which requires that any observable superluminal motion be instantaneous in only one inertial frame (just as it requires that any subluminal motion have zero velocity in only one inertial frame).<sup>37</sup> EPR-type and related quantum correlations are not only nonenergy and noninformation transporting,<sup>38</sup> but also are not even observable, and hence need not be Lorentz invariant.<sup>36</sup> After an EPR-type or related experiment, the receiver can compare notes with the transmitter and thus obtain information concerning the EPR-type and related quantum correlations that had occurred during this experiment but only via communication at speeds allowed by special relativity.<sup>7,31,32,36</sup>

### 5. Can tachyons be detected—if they exist—even indirectly?

Let us consider two cases in point even while recognizing that even the theoretical possibility of tachyons is controversial.<sup>23,25,27</sup> Tachyons if they exist not only cannot be employed for superluminal, let alone supercausal, communication in violation of special relativity,<sup>7,32–35</sup> but also no known method exists even for directly observing them.<sup>28</sup> However if “disappearance” via either traversal or circumvention of the special relativistic speed-of-light barrier of tardyons was to be observed, then this observation might be interpreted as indirect evidence for the existence of tachyons. Such disappearance entails neither communication (superluminal or otherwise) nor direct observation of tachyons. Might an electron or positron tunnel through the light barrier and thence disappear if it could be accelerated at  $\mathcal{A} \geq 7 \times 10^{25} \text{ m/s}^2$  [recall the last two paragraphs of Sec. II C 1]?

If they exist, virtual tachyon-antitachyon pairs might indirectly betray their existence by the modification of vacuum polarization associated with virtual tardyon-antitardyon pairs.<sup>39</sup> (Even though  $u^* = \infty$  for virtual tachyons, the two members of an individual tachyon-antitachyon pair could have  $u \ll u^* = \infty$ , barely exceeding  $c$  and hence perhaps increased likelihood of such betrayal.)

### 6. Trivial superluminal phenomena

Standard special relativity forbids all superluminal motion of energy (including mass) and of information even if causal and not forbidden by an infinite energy cost (with the trivial exception of mutual velocities as large as  $2c$ ).<sup>40</sup> But it allows nonenergy and noninformation transporting superluminal signals.<sup>41</sup> These include, for example, the spot of a rotating searchlight or laser beam sweeping across a sufficiently distant screen, the point of intersection of scissors blades or two rulers that cross each other at an arbitrarily small angle, and that of a falling guillotine blade with its block. Such signals can travel at arbitrarily high velocities.<sup>41</sup> Because these signals are observable, they must obey Lorentz invariance and hence can have infinite velocity in only one inertial frame.<sup>37</sup>

## III. THE ENERGY-TIME UNCERTAINTY PRINCIPLE AND ACCELERATION: UNRUH RADIATION AND THE SCHWINGER EFFECT

The energy-time uncertainty principle can give insights into promotion to real status, that is, materialization, of vir-

tual particles in accelerated systems. We consider the examples of Unruh radiation<sup>8</sup> and the Schwinger effect.<sup>9</sup>

### A. Unruh radiation

#### 1. Unruh radiation via the energy-time uncertainty principle

Consider an Einstein elevator in field-free space with constant rectilinear acceleration  $\mathcal{A}$  from initial speed  $\mathcal{V}_i=0$  to final speed  $\mathcal{V}_f$  relative to its initial instantaneous local inertial frame while traversing the distance  $a$ . In accordance with the equivalence principle a massive real particle in this frame gains (kinetic) energy

$$\Delta E = mc^2[(1 - \mathcal{V}_f^2/c^2)^{-1/2} - 1] \equiv mc^2(\gamma_f - 1) \quad (42a)$$

$$\approx m\langle(1 - \mathcal{V}^2/c^2)^{-1/2}\rangle \mathcal{A}a = m\mathcal{A}a\langle\gamma\rangle \quad (42b)$$

relative to the Einstein elevator. The averages in Eq. (42) are over speeds from  $\mathcal{V}_i=0$  to  $\mathcal{V}_f$  or, equivalently, from  $\gamma_i=1$  to  $\gamma_f$  relative to the Einstein elevator. In accordance with Eqs. (1) and (4) (Refs. 10, 11, 13–15, 17, and 18) a massive virtual particle can with significant probability<sup>10,17,18</sup> be promoted to real status, that is, materialized, if it acquires at least the energy  $\Delta\mathcal{E}=mc^2$  relative to the Einstein elevator in time  $\Delta t \approx \hbar/2\Delta\mathcal{E}$  as measured in the Einstein elevator. For the required energy we have  $\Delta\mathcal{E}=mc^2=\Delta E=mc^2(\gamma_f-1)$ , which implies that  $\gamma_f=2$  and  $\mathcal{V}_f=c(1-\gamma_f^{-2})^{1/2}=\frac{3^{1/2}}{2}c$  or

$$\Delta t = \frac{\mathcal{V}_f}{\mathcal{A}} = \frac{3^{1/2}c}{2\mathcal{A}}. \quad (43)$$

The penalty of increased  $\Delta\mathcal{E}\Delta t$  due to the nonconstancy of  $\mathcal{V}$  is not exorbitant (see Appendix A). In any case our estimate of  $\Delta\mathcal{E}$  is heuristic, probably even more so for virtual particles than for nonrelativistic tunneling. Because this required energy must be accrued within the time  $\Delta t \approx \hbar/2\Delta\mathcal{E}$ , we have

$$\Delta\mathcal{E} = mc^2 \approx \frac{\hbar}{2\Delta t} \approx \frac{\hbar}{2\frac{3^{1/2}c}{2\mathcal{A}}} = \frac{\hbar\mathcal{A}}{3^{1/2}c} \approx \frac{\hbar\mathcal{A}}{c} \approx k_B T, \quad (44)$$

where  $k_B$  is Boltzmann’s constant. The approximation  $\Delta\mathcal{E} \approx \hbar\mathcal{A}/c \approx k_B T$  in Eq. (44) is plausible because the Boltzmann distribution, which implies that  $\Delta\mathcal{E} \approx mc^2 \approx k_B T$ , corresponds to thermodynamic equilibrium. Because the Boltzmann distribution corresponds to thermodynamic equilibrium, it is by far the most probable distribution and hence the distribution of real massive particles into which virtual ones are by far most likely to be materialized for a given  $\mathcal{A}$ .<sup>42</sup> This plausibility argument might be interpreted as an extension of Fermi’s golden rule<sup>43</sup> but applied to transitions to entire probability distributions of states rather than to transitions to states within a given probability distribution.

Similarly, consistent with the special relativistic Doppler effect, a massless real particle (for example, a photon), considered as a wave packet of initial “average” angular frequency  $\langle\omega\rangle$ , accrues (kinetic) energy relative to the Einstein elevator’s initial instantaneous local inertial frame of

$$\Delta E = \hbar\langle\omega\rangle\left(\left[\frac{1 + \mathcal{V}_f/c}{1 - \mathcal{V}_f/c}\right]^{1/2} - 1\right). \quad (45)$$

In accordance with Eqs. (1) and (4) (Refs. 10, 11, 13–15, 17, and 18) a massless virtual particle can with significant

probability<sup>10,17,18</sup> be promoted to real status, that is, materialized, if it accrues at least the energy  $\Delta\mathcal{E}=\hbar\langle\omega\rangle$  relative to the Einstein elevator in time  $\Delta t\approx\hbar/2\Delta\mathcal{E}$  as measured in the Einstein elevator. For the required energy we have

$$\Delta\mathcal{E}\doteq\hbar\langle\omega\rangle=\Delta E=\hbar\langle\omega\rangle\left\{\left[\frac{1+(\mathcal{V}_f/c)}{1-(\mathcal{V}_f/c)}\right]^{1/2}-1\right\}, \quad (46)$$

which implies that

$$\left[\frac{1+(\mathcal{V}_f/c)}{1-(\mathcal{V}_f/c)}\right]^{1/2}=2. \quad (47)$$

Equation (47) yields  $\mathcal{V}_f=3c/5$  and hence

$$\Delta t=\frac{\mathcal{V}_f}{\mathcal{A}}=\frac{3c}{5\mathcal{A}}. \quad (48)$$

Because this required energy must be accrued within the time  $\Delta t\approx\hbar/2\Delta\mathcal{E}$ , we have

$$\Delta\mathcal{E}=\hbar\langle\omega\rangle\approx\frac{\hbar}{2\Delta t}\approx\frac{\hbar}{2\frac{3c}{5\mathcal{A}}}=\frac{5\hbar\mathcal{A}}{6c}\approx\frac{\hbar\mathcal{A}}{c}\approx k_B T. \quad (49)$$

The approximation  $\hbar\langle\omega\rangle\approx\hbar\mathcal{A}/c\approx k_B T$  in Eq. (49) is plausible because the Planck distribution, which implies  $\hbar\langle\omega\rangle\approx k_B T$ , corresponds to thermodynamic equilibrium. Therefore, it is the distribution of real photons into which virtual ones are by far most likely to be materialized for a given  $\mathcal{A}$  [similar to the discussion immediately following Eq. (44)].

By Eqs. (44) and (49) and the discussions immediately following them, the approximate temperature of Unruh radiation is

$$T\approx\frac{\hbar\mathcal{A}}{k_B c}. \quad (50)$$

If we apply Eqs. (19) and (24) with  $V-E=0$  and  $u\rightarrow\mathcal{V}_f$ , and Eq. (44), we obtain for the approximate maximum mass of particles in the Unruh thermal spectrum corresponding to acceleration  $\mathcal{A}$

$$m_{\max}\approx\frac{\hbar\mathcal{A}}{c^3}\approx\frac{k_B T}{c^2}. \quad (51)$$

Based on Eq. (44) and the discussion immediately following it, the first approximation in Eq. (51) is correct to within a small numerical factor. In accordance with the Boltzmann distribution,<sup>42,43</sup> the thermal spectrum of Unruh radiation includes massive particles if  $\mathcal{A}$  is sufficiently large. (Similarly, by the equivalence principle, that is, letting  $\mathcal{A}$  equal gravitational acceleration, massive particles emanate from a sufficiently small Schwarzschild black hole.<sup>44</sup>)

To be materialized a massive ( $m>0$ ) virtual particle must by Eqs. (42)–(44) acquire energy  $\Delta\mathcal{E}\approx m\gamma_f\mathcal{A}a\approx 2m\mathcal{A}a\approx mc^2$  in the time  $\Delta t\approx\hbar/2\Delta\mathcal{E}\approx\mathcal{V}_f/\mathcal{A}\approx(3^{1/2}c/2)/\mathcal{A}=3^{1/2}c/2\mathcal{A}$  while traversing the distance  $a\approx\frac{1}{2}\mathcal{A}(\Delta t)^2\approx\frac{1}{2}\mathcal{A}(3^{1/2}c/2\mathcal{A})^2=3c^2/8\mathcal{A}$  in an accelerational (or gravitational) field  $\mathcal{A}$ . ( $\mathcal{V}_f\approx 3^{1/2}c/2$  is consistent with these Newtonian approximations if we seek only order-of-magnitude accuracy.) By Eq. (4) the probability that it can do so is

$$\begin{aligned} P_{\text{mat},m>0} &\sim e^{-2\Delta\mathcal{E}\Delta t/\hbar} \\ &\approx e^{-[2mc^2(3^{1/2}c/2\mathcal{A})]/\hbar}=e^{-3^{1/2}mc^3/\hbar\mathcal{A}}\approx e^{-mc^3/\hbar\mathcal{A}}. \end{aligned} \quad (52)$$

Similarly, for a massless ( $m=0$ ) virtual particle to be materialized, it must acquire energy  $\Delta\mathcal{E}\approx\hbar\langle\omega\rangle$  in the time  $\Delta t\approx\hbar/2\Delta\mathcal{E}\approx\mathcal{V}_f/\mathcal{A}\approx 3c/5\mathcal{A}$  while traversing the distance  $a\approx c\Delta t\approx c(3c/5\mathcal{A})=3c^2/5\mathcal{A}$  in an accelerational (or gravitational) field  $\mathcal{A}$ . ( $\mathcal{V}_f\approx 3^{1/2}c$  is consistent with these Newtonian approximations for order-of-magnitude accuracy.) The probability it can do so is [see Eq. (4)]

$$\begin{aligned} P_{\text{mat},m=0} &\sim e^{-2\Delta\mathcal{E}\Delta t/\hbar} \\ &\approx e^{-2[\hbar\langle\omega\rangle(3c/5\mathcal{A})]/\hbar}=e^{-6\langle\omega\rangle c/5\mathcal{A}}\approx e^{-\langle\omega\rangle c/\mathcal{A}}. \end{aligned} \quad (53)$$

The exponential forms in Eqs. (52) and (53) become exact for  $P_{\text{mat}}\ll 1$  and are reasonably accurate for  $P_{\text{mat}}\lesssim 1/e$ . Even if the density of virtual particles of a given species is a significant fraction of the Planck density (the total density of all virtual particles in the quantum-mechanical vacuum),  $P_{\text{mat}}$  in Eqs. (52) and (53) is essentially zero if the arguments of the exponents in these equations are greater than about  $10^3$ .

## 2. Unruh radiation, Mach's principle, and thermodynamics

The existence of Unruh radiation disproves an important prediction of Mach's principle.<sup>45</sup> A disk of radius  $R$  rotating at  $\Omega$  rad/s is impinged on by Unruh radiation at temperature  $T=0$  at its center but at temperature  $T\approx\hbar R\Omega^2/k_{BC}$  (for simplicity let  $R\Omega\ll c$ ) at its rim. Contrary to an important prediction of Mach's principle,<sup>45</sup> the disk's absolute rotation (the magnitude, even if not the direction, of its angular-momentum vector) could thus be detected even if it were the only object in the universe.

Consider an otherwise empty universe whose sole content is two identical disks, 1 and 2, that share the same rotational axis. Mach's principle requires that  $\Omega_2=-\Omega_1$ . However Unruh radiation allows the absolute rotation of each of the two disks to be measured individually and hence can break this symmetry. (The trivial case,  $\Omega_1=\Omega_2=0$ , is the only case for which the breaking of this symmetry cannot be demonstrated.)<sup>46</sup>

The Unruh radiation impinges on the disk's rim with a net aberration<sup>47</sup> component of  $\approx R\Omega/c$  against its direction of rotation and hence slows its rotation. (This aberration of Unruh radiation is similar to aberration of raindrops or of starlight.<sup>47</sup>) Thus, in accordance with the first and second laws of thermodynamics, at most only the disk's own rotational kinetic energy can be extracted as work via a heat engine operating between a hot source at the disk's rim and a cold sink at its center. This aberration<sup>47</sup> of Unruh radiation always opposes the acceleration. Thus aberration will result in  $\Omega$  exponentially decaying to zero. In the two-disk example, this decay will be to the trivial case  $\Omega_1=\Omega_2=0$ .

If Unruh radiation, and specifically its aberration,<sup>47</sup> cannot be "turned off," then (accelerated, absolute) rotational motion at constant speed cannot be maintained solely by its own inertia as is (unaccelerated, relative) translational motion at constant speed. Then the associated entropy increase as ordered rotational kinetic energy is thermalized also cannot be turned off. Thus the entropy is required to increase unless already a maximum ( $\Omega=0$ ). We conclude that the second law

of thermodynamics is stronger for rotational motion (entropy must increase) than for translational motion (entropy must either increase or remain constant).

## B. The Schwinger effect

The results obtained in Sec. III A for the materialization of virtual particles in accelerational/gravitational fields apply for other fields. An example is a simple calculation of the minimum electric field strength  $\varepsilon$  required to materialize a virtual electron-positron pair via the Schwinger effect.<sup>9</sup> We apply Eq. (44) except the last step, let  $m_e$  be the electron (or positron) mass and  $q_e$  be the magnitude of the electron (or positron) electric charge, and obtain

$$m = 2m_e = \frac{\Delta \mathcal{E}}{c^2} \approx \frac{\hbar A}{c^3} \approx \frac{\hbar(q_e \varepsilon / m_e)}{c^3}, \quad (54)$$

which implies that  $\varepsilon \approx 2m_e^2 c^3 / \hbar q_e \approx 3 \times 10^{18}$  V/m.

If we apply Eq. (54) to Eq. (52), we obtain for the probability of materialization of a virtual electron-positron pair as a function of  $\varepsilon$

$$\begin{aligned} P_{\text{mat}} &\sim e^{-2mc^3/\hbar A} = e^{-2m_e c^3/\hbar(q_e \varepsilon / m_e)} \\ &= e^{-2m_e^2 c^3/\hbar q_e \varepsilon} \approx \exp\left(-\frac{3 \times 10^{18} \text{ V/m}}{\varepsilon}\right). \end{aligned} \quad (55)$$

The exponential form in Eq. (55) becomes exact for  $P_{\text{mat}} \ll 1$  and is reasonably accurate for  $P_{\text{mat}} \lesssim 1/e$ . Even if the density of virtual electron-positron pairs is a significant fraction of the Planck density,  $P_{\text{mat}}$  is essentially zero for  $\varepsilon \lesssim 10^{-3}(2m_e^2 c^3/\hbar q_e) = 2 \times 10^{-3}(m_e^2 c^3/\hbar q_e) \approx 3 \times 10^{15}$  V/m.

## IV. THE GROUND STATE OF THE ELECTROMAGNETIC FIELD

### A. The ground state in vacuum

Consider a region of vacuum of linear dimension  $\lambda$  where no real photons are present. This region is therefore in the ground state of the electromagnetic field in vacuum. Even in the ground state, the energy-time uncertainty principle implies that there are fluctuations in the electromagnetic field. The typical duration of such fluctuations is  $\Delta t \approx \lambda/c$ . Hence, the magnitude of their energy is  $\Delta \mathcal{E} \approx \hbar/\Delta t \approx \hbar c/\lambda \approx \hbar \nu$ , corresponding to one virtual photon of wavelength  $\lambda$  and frequency  $\nu$  in a volume  $\lambda^3$ . The wavelength and frequency are only roughly defined, with their respective uncertainties not much smaller than their average values. The corresponding (likewise only roughly defined) virtual electromagnetic energy density is  $\rho_{\mathcal{E}} \approx (\hbar c/\lambda)/\lambda^3 = \hbar c/\lambda^4$ .<sup>48</sup> The electromagnetic field can be treated classically if the fluctuating quantum background is negligible, that is, if there are  $\gg 1$  real photons of wavelength  $\approx \lambda$  per volume  $\approx \lambda^3$  and hence a corresponding real electromagnetic energy density  $\gg \hbar c/\lambda^4$ .<sup>48</sup> These results can be derived more rigorously and quantitatively,<sup>48</sup> but the energy-time uncertainty principle gives a more intuitive interpretation. (We could first employ the position-momentum uncertainty principle, which yields  $\Delta p \approx \hbar/\lambda$ , then apply  $\Delta E \approx c \Delta p \approx \hbar c/\lambda = \hbar \nu$ , with subsequent steps the same as before.)

The Planck length  $\ell_{\text{Planck}}$  imposes a lower limit  $\lambda_{\text{min}} \approx \ell_{\text{Planck}}$  on  $\lambda$  and hence an upper limit  $\rho_{\mathcal{E},\text{max}} \approx \rho_{\text{Planck}} \approx \hbar c/\ell_{\text{Planck}}^4$  (the Planck density) on the ground state electromagnetic virtual energy density  $\rho_{\mathcal{E}}$ .<sup>15,27</sup> The virtual energy

density  $\rho_{\mathcal{E},\text{max}} \approx \rho_{\text{Planck}} \approx \hbar c/\ell_{\text{Planck}}^4$  corresponds to one virtual photon of wavelength  $\ell_{\text{Planck}}$  in the ground state of the electromagnetic field and to one virtual black hole per volume  $\ell_{\text{Planck}}^3$ . These estimates hold for real and virtual photons. There cannot be more than one real photon of wavelength  $\ell_{\text{Planck}}$  in a volume  $\ell_{\text{Planck}}^3$  and hence never a real energy density exceeding  $\hbar c/\ell_{\text{Planck}}^4$  because this density would exceed the mass-energy of one real black hole per volume  $\ell_{\text{Planck}}^3$ . Attempting to “squeeze in” more than one real photon per volume  $\ell_{\text{Planck}}^3$  would result in a more massive real black hole. This attempted “squeezing” would cause the real energy density to decrease because the mass-energy of a black hole is proportional to its radius, not to the cube of its radius as would be required to maintain constant density, let alone to increase it. There can never more, let alone much more, than one real photon of wavelength  $\lambda_{\text{min}} \approx \ell_{\text{Planck}}$  in a volume  $\ell_{\text{Planck}}^3$ . Hence, at length scales approaching  $\ell_{\text{Planck}}$ , the fluctuating quantum background can never be made negligible, and the electromagnetic field can never be treated classically, as can be done if  $\lambda \gg \lambda_{\text{min}} \approx \ell_{\text{Planck}}$ .

The density  $\rho_{\mathcal{E},\text{max}}$  is only roughly defined, with its uncertainty being not much smaller than its average value. The magnitude of quantum fluctuations in the metric coefficients over distances  $\approx \ell_{\text{Planck}}$  is comparable to their average values, with virtual black holes forming and reforming on time scales on the order of the Planck time  $\Delta t \approx t_{\text{Planck}} = \ell_{\text{Planck}}/c$ . Hence, concepts such as “length” and “volume” become ambiguous.<sup>15,27</sup>

The prevailing interpretation is that all four fundamental fields (including gravity) and their corresponding quanta become unified at the Planck length scale. Thus the two immediately preceding paragraphs apply to any field and its associated quanta, not only to electromagnetic field and its associated quanta (photons).<sup>15,27</sup>

### B. The ground state in matter

It is interesting to speculate on the ground state of the electromagnetic field in a material of refractive index  $n$ . For simplicity, we assume that the material and hence  $n$  are uniform with  $n \geq 1$ ,<sup>49</sup> and the material is nondispersive so that the phase, group, and signal velocities of light in the material all equal  $c/n$ . Let  $\lambda$  be the wavelength of electromagnetic radiation in vacuum and  $\lambda^*$  be its wavelength in the material. If  $\lambda^*$  is long enough so that the material can be approximated as a continuum, then  $\lambda^* = \lambda/n$ . Hence, the typical duration of quantum fluctuations in a region of linear dimension  $\lambda^*$  is  $\Delta t \approx \lambda^*/(c/n) = n\lambda^*/c$ . The roughly defined energy of such fluctuations is typically  $\Delta \mathcal{E} \approx \hbar/\Delta t \approx \hbar/(n\lambda^*/c) = \hbar c/n\lambda^* \approx \hbar \nu$ , corresponding to one virtual photon of wavelength  $\lambda^*$  and frequency  $\nu$  in a volume  $\lambda^{*3}$ . Hence the corresponding virtual electromagnetic energy density, which is likewise only roughly defined, is  $\rho_{\mathcal{E}} = (\hbar c/n\lambda^*)/\lambda^{*3} = \hbar c/n\lambda^{*4}$ . Because  $\lambda^* = \lambda/n$ , we also have  $\rho_{\mathcal{E}} \approx \hbar c/n\lambda^{*4} = \hbar c/(\lambda^4/n^3) = \hbar c n^3/\lambda^4$ . Thus,  $\rho_{\mathcal{E}} \propto n^3$ . (We require  $\lambda^* \geq \ell_{\text{Planck}}$  or, equivalently,  $\lambda = n\lambda^* \geq n\ell_{\text{Planck}}$ , implying that  $\rho_{\mathcal{E}}$  can never exceed  $\rho_{\text{Planck}}/n$ .) The electromagnetic field can be treated classically if this fluctuating quantum background is negligible, that is, if there are  $\gg 1$  real photons of wavelength  $\lambda^*$  or, equivalently,  $\gg n^3$  in a volume  $\lambda^3$  and hence a corresponding real electromagnetic energy density  $\gg \hbar c/n\lambda^{*4} \approx \hbar c n^3/\lambda^4$ .

If  $\lambda^*$  is not long enough so that the discrete nature of the material can be neglected, then these results are not valid.

However for matter compressed to the Planck density  $\rho_{\text{Planck}}$ , these results hold roughly for  $\lambda^*$  as short as  $\ell_{\text{Planck}}$ .

### C. Emission and absorption

Stimulated emission and absorption are mediated by real photons.<sup>50</sup> Stimulated emission and absorption occur only for an atom in matter, and spontaneous emission occurs whether an atom is in vacuum or in matter.

Could spontaneous emission be stimulated emission that is mediated by virtual photons? After all, during a time interval not longer than  $\Delta t \approx \hbar/\Delta\mathcal{E} \approx \lambda/c$ , an atom should not be able to distinguish between a photon's energy being borrowed or not, that is, between a photon being virtual or real. So, during this brief time interval, might an atom in an excited state accept mediation from a virtual photon as if it were real and hence decay to the ground state (or to a lower excited state)? If a classical vacuum existed with no fluctuating quantum background and hence with no virtual photons (but still with enough quantum mechanics to allow the existence of atoms), then an atom in an excited state in this classical vacuum would never decay.

Moreover, in accordance with Eqs. (1) and (4), it seems plausible, at least heuristically, that virtual photons also mediate temporary spontaneous absorption with typical duration, i.e.,  $e$ -folding time, of  $\Delta t \approx \lambda/c \approx \hbar/\Delta\mathcal{E}$ . Energy conservation forbids virtual processes such as temporary spontaneous absorption from being directly observable, but perhaps there may be some indirect observable effects.

## V. QUANTUM AND THERMAL FLUCTUATIONS

We discuss here three aspects of quantum fluctuations and their relation to thermal fluctuations.

### A. Quantum fluctuations and the Schrödinger equation

Equations (1) and (4) apply even for a particle in an energy eigenstate as given by the Schrödinger equation despite the latter assuming a nonfluctuating classical vacuum and neglecting quantum vacuum fluctuations. A quantum energy fluctuation of magnitude  $\geq \Delta\mathcal{E}$  persisting for time  $\geq \Delta t$ , with corresponding  $P(\geq \Delta\mathcal{E}\Delta t)$  consistent with Eq. (4), is related to fluctuations in the spacetime metric coefficients  $g_{ij}$  with the same probability of occurrence, as given by<sup>15,27</sup>

$$\frac{\Delta\mathcal{E}}{E_{\text{Planck}}} = \frac{\Delta\mathcal{E}}{m_{\text{Planck}}c^2} \sim \frac{\Delta g_{ij}}{g_{ij}} \sim \frac{\ell_{\text{Planck}}}{a}. \quad (56)$$

This fluctuating quantum vacuum contrasts with the nonfluctuating classical vacuum assumed by the Schrödinger equation.

Equations (1) and (4) can be interpreted as allowing temporary energy nonconservation, with energy conservation strictly true in the long-time limit  $\Delta t \rightarrow \infty$ . Even for a particle in an energy eigenstate given by the Schrödinger equation, vacuum fluctuations correspond to energy fluctuations  $\Delta\mathcal{E}$  about the corresponding energy eigenvalue persisting for time  $\Delta t \approx \hbar/2\Delta\mathcal{E}$ . The eigenvalue occurs with certainty only in the limit  $\Delta t \rightarrow \infty$ .<sup>51</sup> [If a particle is not in an energy eigenstate, then it experiences, in addition to  $\Delta\mathcal{E}$ , “standard” quantum-mechanical energy fluctuations, also in accordance with Eq. (1), if it is in a superposition of two or more energy eigenstates,<sup>52</sup> or thermal energy fluctuations if it is in a statistical mixture of two or more energy eigenstates (that is,

with all off-diagonal elements of the density matrix vanishing).<sup>53</sup> (A particle could be partially thermalized, experiencing both types of fluctuations.)]

### B. An imperfect similarity between quantum and thermal fluctuations

There is an at least a superficial similarity between quantum and thermal fluctuations. The Boltzmann distribution  $P(\geq \Delta\mathcal{E}_{\text{th}}) \sim e^{-\Delta\mathcal{E}_{\text{th}}/k_B T}$  ( $\Delta\mathcal{E}_{\text{th}} \equiv \mathcal{E}_{\text{th}} - \mathcal{E}_{\text{th,ground}}$ ) governing the latter can be obtained by substituting  $\Delta\mathcal{E}\Delta t/(1/2\hbar) = 2\Delta\mathcal{E}\Delta t/\hbar \rightarrow \Delta\mathcal{E}_{\text{th}}/k_B T$  into Eq. (4), with  $\Delta\mathcal{E}\Delta t$  and  $\Delta\mathcal{E}_{\text{th}}$  being the fluctuations in the quantum action and thermal energy, respectively. The quantum-vacuum “bath” is the source of quantum action fluctuations of average magnitude  $\langle \Delta\mathcal{E} \rangle \Delta t \approx (\Delta\mathcal{E}_{\text{rms}}\Delta t)_{\text{min}} = \hbar/2$ ,<sup>15,27</sup> a thermal bath is a source of energy fluctuations of average magnitude  $\langle \Delta\mathcal{E}_{\text{th}} \rangle \approx k_B T$ . An average quantum action fluctuation is also an average quantum energy fluctuation of magnitude  $\langle \Delta\mathcal{E} \rangle \approx \hbar/2\Delta t$ . Hence, quantum fluctuations requiring time  $\Delta t$  to occur are more important if  $\langle \Delta\mathcal{E} \rangle > \langle \Delta\mathcal{E}_{\text{th}} \rangle$ , that is, if  $k_B T \leq \hbar/2\Delta t$ , and thermal fluctuations are more important if  $\langle \Delta\mathcal{E}_{\text{th}} \rangle > \langle \Delta\mathcal{E} \rangle$ , that is, if  $k_B T \geq \hbar/2\Delta t$ .<sup>54</sup> (This determines, for example, whether quantum tunneling or classical thermal excitation is more important for surmounting a potential energy barrier that is traversed in  $\Delta t$  during a chemical or nuclear reaction.)

Quantum probabilities entail amplitudes, that is, “preprobabilities,”<sup>55</sup> and hence interference, so any similarity between quantum and thermal fluctuations is imperfect.

### C. Plausibility of Eq. (4) and similarities to the Boltzmann distribution

Equation (4) is not rigorous but is plausible for several reasons.<sup>10</sup> First, the absolute minimum of  $\Delta\mathcal{E}_{\text{rms}}\Delta t$ , that is,  $\hbar/2$ , is the  $e$ -folding scale for quantum-vacuum action fluctuations, as is  $k_B T$  for thermal energy fluctuations. Second, the exponential distribution of  $\Delta\mathcal{E}\Delta t$  in Eq. (4) is the simplest one, as is also true for the exponential distribution of  $\Delta\mathcal{E}_{\text{th}}$  in the Boltzmann distribution. An exponential distribution is the sole one consistent with the only available information being the  $e$ -folding scale and/or average magnitude of the fluctuations<sup>56</sup> and the sole memoryless one.<sup>57</sup> The exponential decreases in probability density with increasing  $\Delta\mathcal{E}\Delta t$  and  $\Delta\mathcal{E}_{\text{th}}$  are sufficiently rapid to be consistent with the correspondence principle, that is, with the respective macroscopic limits quantum mechanics  $\rightarrow$  classical mechanics and statistical mechanics  $\rightarrow$  classical thermodynamics. Third, Eq. (4) is consistent, for example, with the exponential decrease in the penetration probability density with distance into a potential step for which  $V > E$  and with the exponential part of the decrease with distance of forces mediated by massive virtual particles.<sup>24</sup> It is also consistent (in accordance with Ehrenfest's principle) with the quasi-exponential decrease in the probability that the difference between the Feynman path followed by a particle and its most probable (classical) path exceeds a given magnitude. For all of these exponential decreases, the  $e$ -folding scale at least approximately equals  $\hbar/2$ . Fourth, Eq. (4) implies the nonrelativistic results in Eqs. (6)–(9), which agree with those originally derived in Ref. 10 and with nonrelativistic results obtained by other methods.<sup>17–19</sup>

## VI. POSSIBLE PEDAGOGICAL APPLICATIONS

The application of the energy-time uncertainty principle to tunneling by a nonrelativistic particle is common.<sup>1,5,7,10–15,17–19</sup> Students might object that if only  $V - E$  were borrowed, then the particle would have zero kinetic energy and hence zero velocity, so it could not traverse the barrier. The method discussed in Sec. II A accounts also for borrowed kinetic energy and hence minimizes the quantum action fluctuations and could resolve this objection. The pedagogical value of this approach was discussed in Ref. 10.

Typical applications of the energy-time uncertainty principle to the virtual exchange of massive particles usually state that energy  $\approx mc^2$  must be borrowed and that the virtual exchange particle travels at  $\approx c$ .<sup>11–15</sup> Students might object that energy  $mc^2$  corresponds to the particle being at rest, whereas the particle traveling at  $c$  corresponds to infinite energy. In either case, according to Eqs. (4), (18), and (31) [omitting the term containing  $V - E$  in Eq. (18)], the range of the force would be zero. The extension to the relativistic regime in Sec. II B could resolve this objection by applying Eqs. (25)–(30) to find the best possible compromise by minimizing  $\Delta\mathcal{E}\Delta t$ .

Although no experimental violation of the speed-of-light barrier has ever been observed, the two phenomena most commonly credited with protecting its inviolability—the infinite energy cost required for massive particles to reach  $c$  and prevention of violation of causality—seem to protect it incompletely. Maybe even so incompletely that, *prima facie*, breaching it almost seems too easy. The reasons for its almost (but perhaps not absolutely) certain inviolability seem to be deeper and more subtle than that.

We gave a semiquantitative and intuitive explanation of Unruh radiation based on the energy-time uncertainty principle. In contrast, even the simplest quantitative discussions of Unruh radiation based on formal quantum theory<sup>8</sup> are much more difficult. The analogy with materialization of virtual electron-positron pairs via the Schwinger effect might be helpful.

The brief qualitative discussion of the ground state of the electromagnetic field, emission, and absorption in Sec. IV provides another example of an intuitive and heuristic elucidation based on the energy-time uncertainty principle.

The discussion of quantum fluctuations and the Schrödinger equation might provide qualitative insight into how the existence of quantum fluctuations is consistent with the nonfluctuating energy eigenvalues that are solutions to the Schrödinger equation. The Schrödinger approach to formal quantum theory is the easiest one and hence the one that physics students usually learn first. This insight may be enhanced by analogies between quantum and thermal fluctuations, including the plausibility of an exponential probability distribution for the former in (imperfect) analogy with that for the latter.

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## APPENDIX A: $\Delta\mathcal{E}\Delta t$ FOR VARIABLE $u(t)$ AND CONSTANT $u = \langle u(t) \rangle$

Let a particle or signal traverse the distance  $a$  at constant speed  $u = u_0$  in time  $\Delta t = a/u_0$ . If, instead, it were to traverse the distance  $a/2$  at speed  $u_1 < u_0$ , then in order that  $\langle u(t) \rangle = u_0$  and hence  $\Delta t = a/\langle u(t) \rangle = a/u_0$ , it must traverse the remaining distance  $a/2$  at speed  $u_2 > u_0 + (u_0 - u_1) = 2u_0 - u_1 > u_0$  given by

$$u_0 = \frac{a}{\Delta t} = \frac{a}{\Delta t_1 + \Delta t_2} = \frac{a}{\frac{a/2}{u_1} + \frac{a/2}{u_2}} = \frac{1}{\frac{1}{2u_1} + \frac{1}{2u_2}} = \frac{2u_1u_2}{u_1 + u_2}, \quad (\text{A1})$$

that is, at speed

$$u_2 = \frac{u_1u_0}{2u_1 - u_0}. \quad (\text{A2})$$

(For simplicity, we assume that  $u_1$  and  $u_2$  are constant speeds rather than averages of variable ones.) Equations (A1) and (A2) always hold, including for superluminal signals (even if these, although observable in some cases, can only be non-energy and noninformation transporting),<sup>32–35</sup> (hypothetical) superluminal but not supercausal ( $c < u_{\text{signal}} \leq c^2/v_{\text{rel}}$ ) informative signals mutual velocities,<sup>40</sup> tachyons,<sup>25,28–31</sup> and for EPR-type and related quantum correlations that are not even observable and hence need not be Lorentz invariant.<sup>36</sup>

In this appendix we let

$$u_1 = u_0 - \delta u \quad (\delta u \ll u_0, \quad \delta u \ll |u_0 - c|). \quad (\text{A3})$$

Note that  $\delta u = u_1 - u_0$  should not be confused with  $\delta u = |c - u|$  as in Secs. II B and II C. If we apply the binomial theorem through second order, Eq. (A2) for small  $\delta u$  as in Eq. (A3) simplifies to

$$u_2 = u_0 + \delta u + \frac{2(\delta u)^2}{u_0}. \quad (\text{A4})$$

Over the range  $u_1 = u_0 - \delta u < u < u_2 = u_0 + \delta u + 2(\delta u)^2/u_0$ , the borrowed-energy cost can be approximated to second order by

$$\Delta\mathcal{E}_{\delta u} = \Delta\mathcal{E}_{\delta u=0} + \left( \frac{\partial \Delta\mathcal{E}}{\partial u} \right)_{u_0} (u - u_0) + \frac{1}{2} \left( \frac{\partial^2 \Delta\mathcal{E}}{\partial u^2} \right)_{u_0} (u - u_0)^2 \quad (\text{A5a})$$

$$\equiv A + B(u - u_0) + C(u - u_0)^2. \quad (\text{A5b})$$

$A$  is a fixed borrowed-energy cost that is independent of  $u$ , for example,  $V - E$  in Eqs. (2) and (19) and/or  $mc^2$  in Eqs. (19), (22), and (24).

If  $u=u_0=\text{constant}$ , which implies that  $u-u_0=0$ , then

$$\Delta\mathcal{E}\Delta t = (\Delta\mathcal{E}\Delta t)_{\delta u=0} = A\Delta t = A \frac{a}{u_0} = \frac{aA}{u_0}. \quad (\text{A6})$$

If, instead of  $u=u_0=\text{constant}$ ,  $u=u_1=u_0-\delta u$  for the distance  $a/2$  and  $u=u_2=u_0+\delta u+2(\delta u)^2/u_0$  for the remaining distance  $a/2$ , then, applying Eq. (A5) and the binomial theorem through second order, Eq. (A6) is modified to

$$\begin{aligned} (\Delta\mathcal{E}\Delta t)_{\delta u} &= [A - B\delta u + C(\delta u)^2] \frac{\frac{1}{2}a}{u_0 - \delta u} \\ &\quad + \left\{ A + B \left[ \delta u + \frac{2(\delta u)^2}{u_0} \right] + C(\delta u)^2 \right\} \\ &\quad \times \frac{\frac{1}{2}a}{u_0 + \delta u + \frac{2(\delta u)^2}{u_0}} \end{aligned} \quad (\text{A7a})$$

$$\begin{aligned} &= \frac{a}{2} \left\{ [A - B\delta u + C(\delta u)^2] \left[ u_0 \left( 1 - \frac{\delta u}{u_0} \right) \right]^{-1} \right. \\ &\quad + \left[ A + B\delta u + \frac{2B(\delta u)^2}{u_0} + C(\delta u)^2 \right] \\ &\quad \times \left. \left[ u_0 \left( 1 + \frac{\delta u}{u_0} + 2\frac{(\delta u)^2}{u_0^2} \right) \right]^{-1} \right\} \end{aligned} \quad (\text{A7b})$$

$$\begin{aligned} &= \frac{a}{2u_0} \left\{ [A - B\delta u + C(\delta u)^2] \left[ 1 + \frac{\delta u}{u_0} + \frac{(\delta u)^2}{u_0^2} \right] \right. \\ &\quad + \left[ A + B\delta u + \frac{2B(\delta u)^2}{u_0} + C(\delta u)^2 \right] \\ &\quad \times \left. \left[ 1 - \frac{\delta u}{u_0} - 2\frac{(\delta u)^2}{u_0^2} + \frac{(\delta u)^2}{u_0^2} \right] \right\} \end{aligned} \quad (\text{A7c})$$

$$\begin{aligned} &= \frac{aA}{u_0} + \frac{aC(\delta u)^2}{u_0} \\ &= (\Delta\mathcal{E}\Delta t)_{\delta u=0} + \frac{a}{2u_0} \left( \frac{\partial^2 \Delta\mathcal{E}}{\partial u^2} \right)_{u_0} (\delta u)^2. \end{aligned} \quad (\text{A7d})$$

The first (fixed-borrowed-energy-cost) term in the final result in Eq. (A7d) remains unmodified from Eq. (A6). Only the second (kinetic energy) term is extra, that is, not present in Eq. (A6). This modification results in  $(\Delta\mathcal{E}\Delta t)_{\delta u} > (\Delta\mathcal{E}\Delta t)_{\delta u=0}$  if  $C > 0$ —as is always the case for tunneling of a massive particle with  $0 \leq u < c$  (where  $\Delta\mathcal{E}$  increases monotonically with increasing  $u$ ). By Eq. (4), for given  $V-E$  and  $a$ , tunneling is easiest and most probable if  $\Delta\mathcal{E}\Delta t$  is minimized. Hence, for tunneling of a massive particle with  $0 \leq u < c$ , constant  $u=u_0$ , that is, constant  $\delta u=0$ , is more probable than any nonconstant  $u(t)$  whose average  $\langle u(t) \rangle$  equals  $u_0$ .

In contrast, in the tachyonic regime  $u > c$ ,  $\Delta\mathcal{E}$  decreases monotonically with increasing  $u$ , so  $C < 0$  could correspond to  $\Delta\mathcal{E}\Delta t$  being minimized there. Thus, it might be reasonable that the variability of  $u$  does not exact an exorbitant  $(\Delta\mathcal{E}\Delta t)_{\delta u}$  penalty for tachyonic tunneling. In the tachyonic regime this penalty might even be negative. The  $\Delta\mathcal{E}\Delta t$  penalty for nonconstancy of  $\mathcal{V}$  in Sec. III A is positive, albeit typically not exorbitant.

Massless virtual particles move at fixed speed  $u=c$ , so minimization of  $\Delta\mathcal{E}\Delta t$  with respect to  $u$ , or with respect to constancy of  $u$ , is not applicable for them. Also, for fixed  $\Delta\mathcal{E}$ ,  $\Delta\mathcal{E}\Delta t$  is neutral with respect to constancy of  $\Delta\mathcal{E}=\hbar\omega$ , for example,  $\Delta\mathcal{E}\Delta t=\hbar\omega\Delta t=a\hbar\omega/c$  given either constant  $\omega=\omega_0$  through distance  $a$  or  $\omega_1=\omega_0-\delta\omega$  through distance  $a/2$  and  $\omega_2=\omega_0+\delta\omega$  through the remaining distance  $a/2$  ( $\delta\omega < \omega$ ).

## APPENDIX B: ON-MASS-SHELL ON THE AVERAGE?

We give a heuristic plausibility argument of our implicit assumption that even if only on the average, borrowed kinetic energy is on-mass-shell, that is,  $K=mc^2[(1-u^2/c^2)^{-1/2}-1]$ . For our calculated values of  $\Delta\mathcal{E}\Delta t [= \Delta\mathcal{E}(a/u)]$  to be valid, even if only on the average, this assumption must be valid. Let the deviation from the mass-shell be  $D \equiv K - mc^2[(1-u^2/c^2)^{-1/2}-1]$ . In classical mechanics,  $D=0$  exactly with certainty in all cases. Only in classical mechanics are both kinetic energy and momentum simultaneously always on-mass-shell. In contrast, in quantum mechanics, even for real particles, the absolute value of the momentum has the corresponding on-mass-shell (classical) value only on the average if the kinetic energy is fixed and has exactly the corresponding on-mass-shell (classical) value with nonzero probability density but with probability of measure zero.<sup>58–60</sup> (If, instead, the absolute value of the momentum is fixed so that the kinetic energy fluctuates, then switch “absolute value of the momentum” and “kinetic energy” in the immediately preceding sentence.<sup>58,59</sup>  $D$  is almost always expressed in terms of fluctuating kinetic energy given fixed magnitude of momentum.<sup>59</sup> However there seems no obvious argument against the reverse expression.<sup>58,59</sup>) For quantum-mechanical virtual particles, a probability even of measure zero that  $D=0$  exactly may be forbidden, that is, this probability may be required to be strictly zero.<sup>60–62</sup> [That  $D=0$  exactly occurs with probability of strictly zero<sup>60–62</sup>—in contrast to probability of measure zero—is more strongly stated as being necessarily true for “totally virtual” particles, whose total energy is borrowed,<sup>61,62</sup> than for borrowed kinetic energy of pre-existing real particles (as in tunneling, especially nonrelativistic tunneling, where the borrowed energy is tiny compared to the pretunneling total energy  $\epsilon=mc^2+mv^2/2$ ).<sup>63</sup>]

For quantum-mechanical real particles it is rigorously true that  $\langle D \rangle = 0$  due to the offsetting contributions of quantum fluctuations with  $D > 0$  and  $D < 0$  even if  $D=0$  occurs with probability as small as measure zero (but even then is still a possible event).<sup>58–60</sup> So, it might be reasonable to assume that for quantum-mechanical virtual particles, offsetting contributions of quantum fluctuations with  $D > 0$  and  $D < 0$  yield  $\langle D \rangle = 0$  even if  $D=0$  exactly occurs with probability of strictly zero rather than with measure zero (and thus is an impossible event).<sup>60–63</sup> This assumption seems to be in accord with the classical limit, that is, with the average of all possible quantum-mechanical behaviors yielding classical behavior. The “most classical” quantum fluctuation or virtual particle is on-mass-shell on the average ( $\langle D \rangle = 0$ ) despite the fact that even if  $\langle D \rangle = 0$ ,  $D=0$  never occurs.<sup>60–63</sup> (This behavior may occur for virtual photons.<sup>64</sup>) This assumption is consistent, for example, with Ehrenfest’s theorem<sup>20</sup> and with constructive interference between Feynman paths being maximized at the classical path and becoming negligible beyond a few de Broglie wavelengths away from the classical

path.<sup>26</sup> It also is consistent with the nonrelativistic results derived by Cohen<sup>10</sup> and evaluated by Hagmann,<sup>18</sup> which in turn are consistent with those obtained by numerous other methods.<sup>17</sup>

<sup>1</sup>For examples of opposing textbook interpretations of the aspects of the energy-time uncertainty principle most relevant to this paper, compare the viewpoint that energy is strictly conserved in quantum mechanics given in Ref. 36, pp. 78–80, to the viewpoint that virtual processes can temporarily violate energy conservation given in R. Shankar, *Principles of Quantum Mechanics*, 2nd ed. (Plenum, New York, 1994), Sec. 9.5. Discussions of aspects of the latter interpretation can be found in Refs. 31 and 65–67. Discussions from a more or less neutral viewpoint are given in Ref. 68.

<sup>2</sup>We assume that virtual processes can temporarily violate energy conservation. In this interpretation energy conservation is absolutely valid in the long term, but the energy-time uncertainty principle is construed as allowing temporary energy fluctuations about a conserved long-term average value.

<sup>3</sup>D. H. Kobe and V. C. Aguilera-Navarro, “Derivation of the energy-time uncertainty relation,” *Phys. Rev. A* **50**, 933–938 (1994), and references cited therein.

<sup>4</sup>The energy-time uncertainty principle is more difficult to interpret than the position-momentum uncertainty principle primarily because position and momentum (and also energy) are operators in quantum mechanics, whereas the status of time in quantum mechanics is a more difficult issue. This point was emphasized by Dr. D. H. Kobe private communication (2007). The status of time quantum mechanics, and related issues, are addressed in Ref. 3. Dr. Kobe also pointed out to me that the usual definition of the uncertainty product requires some refinement in order to be completely accurate, even though it is sufficiently accurate for our purposes in this present paper. See, for example, J. Hilgevoord and J. B. M. Uffink, “More uncertainty about the uncertainty principle,” *Eur. J. Phys.* **6**, 165–170 (1985); and J. B. M. Uffink, and J. Hilgevoord, “Uncertainty principle and uncertainty relations,” *Found Phys.* **15**, 925–944 (1985), which however, considers the position-momentum rather than the energy-time uncertainty principle.

<sup>5</sup>If relativistic effects are important, then “energy” should be interpreted accordingly. See Refs. 69–73.

<sup>6</sup>It has been stated that the relations  $E=h\nu$  and  $p=h/\lambda$  cannot be understood in terms of anything more basic. See L. Brillouin, *Relativity Reexamined* (Academic, New York, 1970), Chaps. 1 and 3, and especially Secs. 3.3 and 3.4, and p. 34. We do not know why, for example, Planck’s constant  $h$  has its measured value, nor why it is  $h$  rather than another dimensionally equivalent constant that “binds”  $p$  and  $\lambda$ , and  $E$  and  $\nu$ . (Dimensional equivalence need not imply physical equivalence; for example, entropy and heat capacity are dimensionally but not physically equivalent.) However, Brillouin’s statement may be too strong because the proportionalities  $p \propto 1/\lambda$  and  $E \propto \nu$  are suggested by special relativity. See Ref. 69, p. 13, Exercise 2.6, p. 58, Sec. 6.8, and Exercise 6.20. See also Ref. 70, pp. 2 and 62–63, Secs. 32–33, and Exercises 24 and 25 of Chap. V. In general relativity the proportionality  $E \propto \nu$  ensures consistency of gravitational time dilation and energy conservation during gravitational frequency shift. See Ref. 69, Sec. 1.16, Chap. 9, Secs. 11.6 and 12.2, and Exercises 1.8–1.11, 6.20, and 9.1–9.4; An excellent summary of these points is in P. Fong, *Elementary Quantum Mechanics, Expanded Edition*, (World Scientific, Singapore, 2005), pp. 26–47; Also, see relevant references cited therein. A relevant summary of de Broglie’s viewpoints concerning  $h$  is in L. de Broglie, *New Perspectives in Physics* (Basic Books, New York, 1962), translated by A. J. Pomerans.

<sup>7</sup>P. C. W. Davies, “Quantum tunneling time,” *Am. J. Phys.* **73**, 23–27 (2005).

<sup>8</sup>P. M. Alsing and P. W. Milonni, “Simplified derivation of the Hawking–Unruh temperature for an accelerated observer in vacuum,” *Am. J. Phys.* **72**, 1524–1529 (2004), and references cited therein. Also, V. F. Mukhanov and S. Winitzki, *Introduction to Quantum Effects of Gravity* (Cambridge U. P., Cambridge, 2007), p. 12 and Chap. 8.

<sup>9</sup>F. Mukhanov and S. Winitzki, *Introduction to Quantum Effects of Gravity* (Cambridge U. P., Cambridge, 2007), pp. 11–12 and the solution to Exercise 1.5.

<sup>10</sup>See Bernard L. Cohen, “A simple treatment of potential barrier penetration,” *Am. J. Phys.* **33**, 97–98 (1965), and the reference cited therein.

<sup>11</sup>H. M. Georgi, in *The New Physics*, edited by P. Davies (Cambridge U. P., Cambridge, 1989).

<sup>12</sup>J. Taylor, in *The New Physics*, edited by P. Davies (Cambridge U. P., Cambridge, 1989).

<sup>13</sup>R. Adler, in *The New Physics for the Twenty-First Century*, edited by G. Fraser (Cambridge U. P., Cambridge, 2006).

<sup>14</sup>Michael B. Green, in *The New Physics for the Twenty-First Century*, edited by G. Fraser (Cambridge U. P., Cambridge, 2006).

<sup>15</sup>Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler, *Gravitation* (Freeman, New York, 1973), Sec. 43.4. In this present paper, we can neglect the “extra” term in the generalized uncertainty principle as discussed in Ref. 13, Sec. 2.9.

<sup>16</sup>See Leonard I. Schiff, *Quantum Mechanics*, 3rd ed. (McGraw-Hill, New York, 1968), Sec. 17.

<sup>17</sup>See Rolf Landauer, “Barrier traversal time,” *Nature (London)* **341**, 567–568 (1989); M. Büttiker and R. Landauer, “Traversal time for tunneling,” *Phys. Rev. Lett.* **49**, 1739–1742 (1982); Markus Büttiker and Rolf Landauer, “Traversal time for tunneling,” *IBM J. Res. Dev.* **30**, 451–454 (1986) and references cited therein. These authors often interpret the (nonrelativistic) result  $\Delta t = a[m/2(V-E)]^{1/2}$  as a (nonrelativistic) tunneling particle’s barrier-traversal time as we do. Sometimes these authors interpret  $\Delta t$  as something more like a (nonrelativistic) tunneling particle’s interaction time with the barrier.

<sup>18</sup>Numerous approaches agree with the results in Ref. 10 for the tunneling time, at least for barriers that are not highly transparent. See Mark J. Hagmann, “Transit time for quantum tunneling,” *Solid State Commun.* **82**, 867–870 (1992); Mark J. Hagmann, “Quantum tunneling times: A new solution compared to 12 other methods,” *Int. J. Quantum Chem.* **44**, 299–309 (1992); Mark J. Hagmann, “Distribution of times for barrier traversal caused by energy fluctuations,” *J. Appl. Phys.* **74**, 7302–7305 (1993); Mark J. Hagmann, “Effects of the finite duration of quantum tunneling in laser-assisted scanning tunneling microscopy,” *Int. J. Quantum Chem.* **52**, 271–282 (1994); Mark J. Hagmann, “Reduced effects of laser illumination due to the finite duration of quantum tunneling,” *J. Vac. Sci. Technol. B* **12**, 3191–3195 (1994).

<sup>19</sup>Many approaches agree that the nonrelativistic results in Ref. 10 require barriers that are not highly transparent. See Refs. 17 and 18. Also, reflection from a totally opaque potential step, especially of a wide wave packet, manifests major differences in behavior from tunneling through even a nearly opaque potential barrier. If the second strong inequality in Eq. (12) is reversed as for the case of the time delay upon reflection from an infinitely wide potential step for which  $V > E$ , and  $a_{\text{efold}}$  is the  $e$ -folding penetration distance of  $|\Psi|^2$  into a wide ( $\Delta x \gg a_{\text{efold}}$ ) wave packet, then even at midreflection only a small part of the wave packet enters the step; most of it remains outside (as incident and reflected waves). See Refs. 20 and 68.

<sup>20</sup>Claude Cohen-Tannoudji, Bernard Diu, and Frank Laloe, *Quantum Mechanics* (Wiley, New York, 1977), Complement J<sub>1</sub>.

<sup>21</sup>As discussed in Ref. 20, Complement H<sub>1</sub>, the  $e$ -folding distance of  $|\Psi|^2$  is  $a_{\text{efold}} = \hbar/[8m(V-E)]^{1/2}$  into an infinitely wide potential step for which  $V > E = \frac{1}{2}mv^2$ ;  $+v$  ( $-v$ ) is the particle velocity outside the step before (after) reflection. Because the decay is exponential,  $a_{\text{efold}}$  is also the average penetration into the step of the small part of a wide ( $\Delta x \gg a_{\text{efold}}$ ) wave packet that penetrates. The round-trip distance within the step traversed by this part of the wave packet upon reflection is  $2a_{\text{efold}}$ . As shown in Ref. 20, the time delay upon reflection is  $\tau = (\hbar/v)[2/m(V-E)]^{1/2}$ . Hence, for this part of the wave packet, the effective speed within the step during the reflection process is  $u^* = 2a_{\text{efold}}/\tau = v/2$ . This result for  $u^*$  disagrees with our nonrelativistic result in Eq. (6). However reflection from a potential step of infinite width manifests major differences from tunneling through a potential barrier for which  $V > E$  of finite width (or possibly of infinitesimal width if infinitely tall). In tunneling the entire wave packet (not just a small part) must traverse the entire barrier. Also as discussed in Ref. 20, Complement A<sub>III</sub>, if the second strong inequalities in Eq. (10) and/or Eq. (12) are reversed, then the probability distribution of the (nonrelativistic) velocity of a particle of mass  $m$  in a kinetic energy eigenstate with eigenvalue  $K$  has, during its interaction with a barrier (or well) of width  $a$ , a large relative dispersion about the rms (classical) value of  $(2K/m)^{1/2}$ .

<sup>22</sup>See Ref. 10, p. 97, Robert Gomer, *Field Emission and Field Ionization* (Harvard U. P., Cambridge, MA, 1961), pp. 1–8, Phillip H. Bligh, “Note on the uncertainty principle and barrier penetration,” *Am. J. Phys.* **48**, 337–338 (1974).

<sup>23</sup>The signal velocity is defined as “including all points of nonanalyticity,

- that is, new information that is not already foretold in an earlier portion of the waveform." It is not "the velocity of propagation of the half-the-peak-intensity point on the leading part of the pulse" (Ref. 31). Even in cases where the phase velocity, the group velocity and/or the energy transport velocity exceed  $c$ , the signal velocity thus defined does not exceed  $c$ , and hence causality is not violated regardless of the motion of transmitter and/or receiver. See Ref. 31, Chaps. 1–4; Raymond Y. Chiao, Paul G. Kwiat, and Aephraim M. Steinberg, "Faster than light?" *Sci. Am.* **269**(2), 52–60 (1993); A. M. Steinberg, P. G. Kwiat, and R. Y. Chiao, "Measurement of the single-photon tunneling time," *Phys. Rev. Lett.* **71**, 708–711 (1993); Raymond Y. Chiao, "Superluminal (but causal) propagation of wave packets in transparent media with inverted atomic populations," *Phys. Rev. A* **48**, R34–R37 (1993); and D. R. Solli, C. F. McCormick, C. Ropers, J. J. Morehead, R. Y. Chiao, and J. M. Hickmann, "Demonstration of superluminal effects in an absorptionless, nonreflective system," *Phys. Rev. Lett.* **91**, 143906–1–4 (2003); Another viewpoint is in Edward Gerjuoy and Andrew M. Sessler, "Popper's experiment and communication," *Am. J. Phys.* **74**, 643–648 (2006).
- <sup>24</sup> See Chris Isham, in *The New Physics*, edited by Paul Davies (Cambridge U.P., Cambridge, 1989). The inverse-square part of this decrease in forces with distance is a purely geometrical consequence of (Euclidean) space being three-dimensional.
- <sup>25</sup> The tachyonic regime is discussed in Gerald Feinberg, "Particles that go faster than light," *Sci. Am.* **222**(2), 68–77 (1970); G. Feinberg, "Possibility of faster-than-light particles," *Phys. Rev.* **159**, 1089–1105 (1967); Oleksa-Myron Bilaniuk and E. C. George Sudarshan, "Particles beyond the light barrier," *Phys. Today* **22**(5), 43–52 (1969); J. D. Elder, G. L. Trigg, O. M. Bilaniuk, and E. C. G. Sudarshan, "Tachyons revisited," *Phys. Today* **22**(10), 15 and 79 (1969); and O. M. Bilaniuk, S. L. Brown, B. DeWitt, W. A. Newcomb, M. Sachs, G. Sudarshan, and S. Yoshikawa, "More about tachyons," *Phys. Today* **22**(12), 47–52 (1969). See also references cited in these works.
- <sup>26</sup> R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, New York, 1965).
- <sup>27</sup> See Ref. 15, Chap. 43, and Secs. 44.2–44.3. For more recent work see W. A. Christiansen, Y. Jack Ng, and H. van Dam, "Probing spacetime foam with extragalactic sources," *Phys. Rev. Lett.* **96**, 051301–1–4 (2006); **98**, 259903(E) (2007); Also, Y. Jack Ng, "Quantum foam and quantum gravity phenomenology," in *Planck Scale Effects in Astrophysics and Cosmology*, edited by G. Amelino-Camelia and J. Kowalski-Glikman (Springer, Berlin, 2005); and Y. Jack Ng and H. van Dam, "Spacetime foam: Holographic principle, and black hole quantum computers," *Int. J. Mod. Phys. A* **20**, 1328–1335 (2005); More qualitative discussions are given in Refs. 13 and 14. A recent review is given in Lee Smolin, "Quantum gravity faces reality," *Phys. Today* **59** (11), 44–48 (2006). In this present paper, we consider mainly distances and time intervals much larger than the Planck dimensions, with only one brief conjecture (in the next-to-last paragraph of Sec. II C 1) concerning hypothetical distances and time intervals smaller than the Planck dimensions. Hence, we can neglect the "extra" term in the uncertainty principle discussed in Ref. 13, Sec. 2.9. An alternative viewpoint of the quantum vacuum is summarized in C. Lanczos, *The Einstein Decade (1905–1915)* (Academic, New York, 1974), pp. 31–35; with a more detailed discussion in C. Lanczos, "Vector potential in Riemannian space," *Found. Phys.* **4**, 137–147 (1974). Recent work suggesting a possible breakdown of special relativity that is related to quantum field theory is reviewed in Ron Cowen, "Neutrino data hint at need for revised theories," *Science News* **178** (2), 9 (2010).
- <sup>28</sup> Catherine Asaro, "Complex speeds and special relativity," *Am. J. Phys.* **64**, 421–429 (1996), and references cited therein.
- <sup>29</sup> Nick Herbert, *Faster Than Light* (Penguin, New York, 1988), Chap. 7.
- <sup>30</sup> See Gordon Kane, *Modern Elementary Particle Physics* (Addison-Wesley, Reading, MA, 1993). The laws of lepton-number and baryon-number conservation are very good approximations but do allow permanent violations in rare and/or extreme circumstances. In contrast, the law of energy conservation is absolutely valid in the long-time limit  $\Delta t \rightarrow \infty$ , even if the energy-time uncertainty principle is interpreted as allowing temporary energy fluctuations about an absolutely conserved long-term average value.
- <sup>31</sup> P. W. Milonni, *Fast Light, Slow Light, and Left-Handed Light* (Institute of Physics, Bristol, 2005), p. 63, and Secs. 3.1.1, 3.2.1, and 3.8.
- <sup>32</sup> Reference 69, Sec. 2.10 (especially the first paragraph). Also, Ref. 70, Item (ix) of Sec. 7 (especially the first paragraph).
- <sup>33</sup> Richard C. Tolman, *The Theory of the Relativity of Motion* (Dover, Mineola, NY, 2004), Sec. 52.
- <sup>34</sup> Note that the maximum signal velocity consistent with causality,  $u_{\text{signal,max}} = c^2/v_{\text{rel}}$  ( $v_{\text{rel}}$  is the relative velocity between the transmitter and receiver), is also the de Broglie wave speed.
- <sup>35</sup> We mention that extrasensory perception if it exists could propagate superluminally as fast as  $u_{\text{signal,max}} = c^2/v_{\text{rel}}$  and still be a causal phenomenon. The possibility of some phenomena being supercausal and not merely superluminal if they exist is discussed in *Frontiers of Time: Retrocausality—Experiment and Theory*, edited by Daniel P. Sheehan (American Institute of Physics, Melville, NY, 2006) (By contrast, according to standard special relativity, even such unproven phenomena if they exist cannot propagate faster than  $c$ , even though this need not violate causality. See Ref. 69, p. 27, and Ref. 70, p. 16.) For general discussions of both sides of the issue, see for example: Dr. E. Mitchell with D. Williams, *The Way of the Explorer*, Revised Edition (Career Press, Franklin Lakes, NJ, 2008); and R. Matthews, "Quantum entanglement," BBC Knowledge, 75–79, especially p. 77 (May–June 2009); versus M. Kaku, *Physics of the Impossible* (Doubleday, New York, 2008), Chaps. 5, 6, and 15, with Notes and references for Chaps. 5 and 6 on pp. 308–310; and J. Randi, *Flim-Flam!* (Prometheus Books, Amherst, NY, 1982).
- <sup>36</sup> D. J. Griffiths, *Introduction to Quantum Mechanics*, 2nd ed. (Pearson-Prentice Hall, Upper Saddle River, NJ, 2005), Sec. 3.5.3.
- <sup>37</sup> Reference 69; Sec. 2.10 (especially the seventh, eighth, and ninth paragraphs), Sec. 2.11, and the references cited therein; and Exercises 2.6, 2.7, and 2.12 on pp. 58–59. Also, Ref. 70, Sec. 7 [especially the first, third, and fourth paragraphs of Item (ix)] and Exercises 5–7 and 15 of Chap. I on pp. 21–23.
- <sup>38</sup> The possibility of superluminal but not supercausal signals is discussed in Mitchell J. Feigenbaum, "The theory of relativity—Galileo's child," arXiv:0806.1234v1, and reviewed by Mark Buchanan, "Lights out on Einstein's relativity," *New Sci.* **199** (2680), 28–31 (2008).
- <sup>39</sup> See Refs. 11 and 12.
- <sup>40</sup> Reference 69, Sec. 3.6 (especially the last paragraph). Also, Ref. 70, p. 18, and Exercise 5 of Chap. II on p. 36.
- <sup>41</sup> Reference 69, Secs. 2.9–2.11 (especially Fig. 2.6 on p. 50, the first complete paragraph on p. 51, and the third complete paragraph on p. 56) and Exercises 2.6 and 2.7 on pp. 58–59. Also, Ref. 70, p. 18, Exercises 6, 7, and 15 of Chap. I and Sec. 32.
- <sup>42</sup> An alternative qualitative description of the thermal nature of Unruh radiation is given in Lee Smolin, *Three Roads to Quantum Gravity* (Basic Books, New York, 2001), Chap. 6. In Chap. 7 this description is extended to Hawking radiation in accordance with the equivalence principle.
- <sup>43</sup> Transition probabilities are discussed in Ref. 20, Chap. XIII (especially Sec. C.3.b). See also Ref. 67, Sec. 10.1.
- <sup>44</sup> See S. W. Hawking, "Black hole thermodynamics," *Phys. Rev. D* **13**, 191–198 (1976); S. W. Hawking, "The quantum mechanics of black holes," *Sci. Am.* **236**(1), 34–40 (1977) and references cited therein. See also F. Mukhanov and S. Winitzki, *Introduction to Quantum Effects of Gravity* (Cambridge U. P., Cambridge, 2007) and Ref. 13.
- <sup>45</sup> A qualitative discussion is given in Ref. 42.
- <sup>46</sup> Reference 69, p. 8. In a few well-chosen words: "... while in general relativity all matter, including its motion, undoubtedly affects local inertial behavior, it appears not entirely to cause it."
- <sup>47</sup> Reference 69; Exercise 2.1 on p. 58; Sec. 4.4; Exercises 4.5, 4.6, 4.10, 4.13, 4.14, and 4.16 (especially Exercise 4.10); and p. 104. Also, Ref. 70; pp. 1, 42–43, and 65; and Exercises 9, 6, and 17 of Chap. III; and Ref. 33, Sec. 55.
- <sup>48</sup> J. J. Sakurai, *Advanced Quantum Mechanics* (Addison-Wesley, New York, 1967), pp. 32–37.
- <sup>49</sup> Transparency can be induced even if  $n$  is considerably greater than one. See Marlan O. Scully and M. Suhail Zubary, *Quantum Optics* (Cambridge U. P., Cambridge, 1997), pp. 220–221 and Secs. 7.3 and 7.6.
- <sup>50</sup> Richard P. Feynman, Robert B. Leighton, and Matthew Sands, *The Feynman Lectures on Physics* (Addison-Wesley, Reading, MA, 1963), Vol. 1, Secs. 42–5.
- <sup>51</sup> See Refs. 1–5.
- <sup>52</sup> See, for example, Ref. 20, Sec. D.2.e of Chap. III; Ref. 65, pp. 78–80 and Secs. 5.6 and 5.8; and Ref. 68, p. 137 and Sec. 13 of Chap. VIII.
- <sup>53</sup> Ralph Baierlein, *Atoms and Information Theory* (Freeman, San Francisco, 1971), Chap. 12.
- <sup>54</sup> L. D. Landau and E. M. Lifshitz, *Statistical Physics*, 3rd ed. (Pergamon, Oxford, 1989), pp. 333–334; and Ref. 66, Secs. 1, 16, and 44.
- <sup>55</sup> Richard W. Hamming, *The Art of Probability for Scientists and Engineers* (Perseus Books, Cambridge, MA, 1991), pp. 2–3 and Secs. 8.12–8.13. See especially p. 3 and Sec. 8.13.

- <sup>56</sup> See, for example, Ref. 53, Chaps. 3 and 4.
- <sup>57</sup> Sheldon Ross, *Stochastic Processes* (Wiley, New York, 1996), 2nd ed., Sec. 1.6. It is often stated that the exponential distribution is the only memoryless continuous probability distribution and that the geometric distribution is the only memoryless discrete probability distribution. However the geometric distribution can be expressed in exponential form, i.e., as a discrete exponential distribution. Aspects of this point are discussed in Saeed Ghahramani, *Fundamentals of Probability with Stochastic Processes*, 3rd ed. (Pearson-Prentice-Hall, Upper Saddle River, NJ, 2005).
- <sup>58</sup> See Ref. 20, Complement A<sub>III</sub> (see especially Part 1) and the associated discussions in Ref. 19.
- <sup>59</sup> Donald H. Kobe, private communications (2007).
- <sup>60</sup> A probability of measure zero corresponds roughly to one chance in  $\infty$ , for example, the probability of obtaining exactly five given a uniform probability density over the real numbers from zero to ten. A probability of strictly zero corresponds to zero chance in  $\infty$ , for example, the probability of obtaining exactly 15 given the same, or the probability of 100 cm annual precipitation at a weather station whose annual precipitation is 50 cm in half of the years and 150 cm in the other half—the average of 100 cm never occurs in any year. For helpful discussions see John E. Freund, *A Modern Introduction to Mathematics* (Prentice-Hall, Englewood Cliffs, NJ, 1956), Chap. 22; *The VNR Concise Encyclopedia of Mathematics*, edited by W. Gellert, S. Gottwald, M. Hellwich, H. Kästner, and H. Küstner (Van Nostrand Reinhold, New York, 1989), 2nd ed., Chap. 14; Isaac Asimov, *On Numbers* (Bell, New York, 1977), Chap. 5; and John D. Barrow, *The Infinite Book* (Vintage Books, New York, 2005), especially Chap. 4 and pp. 67–76.
- <sup>61</sup> See Martinus Veltman, *Facts and Mysteries in Elementary Particle Physics* (World Scientific, New York, 2003), Chap. 4 and 9.
- <sup>62</sup> Roger Penrose, *The Road to Reality* (Knopf, New York, 2006), Sec. 26.8.
- <sup>63</sup> However, see Ref. 61, p. 252.
- <sup>64</sup> Reference 67, Sec. 10.6 (especially p. 306).
- <sup>65</sup> J. J. Sakurai, *Modern Quantum Mechanics*, revised ed. (Addison-Wesley, Reading, MA, 1994), pp. 78–80 and Secs. 5.6 and 5.8.
- <sup>66</sup> L. D. Landau and E. M. Lifshitz, *Quantum Mechanics*, 2nd revised ed. (Butterworth-Heinemann, Oxford, 1999), Sec. 44.
- <sup>67</sup> E. M. Henley and A. Garcia, *Subatomic Physics*, 3rd ed. (World Scientific, Hackensack, NJ, 2007), Sec. 5.8, pp. 284–286, and 3.10, and Problem 10.31 on p. 329.
- <sup>68</sup> A. Messiah, *Quantum Mechanics* (Wiley, New York, 1958).
- <sup>69</sup> W. Rindler, *Relativity: Special, General, and Cosmological*, 2nd ed. (Oxford U. P., Oxford, 2006), Sec. 6.2–6.3 (especially Sec. 6.3). See also the reference cited in Sec. 6.2. In 1905, Einstein could prove only “that energy *contributes* to mass, without (necessarily) causing *all* of it”—it was still logically possible to assume that particles had inviolable “core” masses. “To equate *all* mass with energy (in 1905, and even in the subsequent early years of special relativity) required an act of aesthetic faith, very characteristic of Einstein.” Interpretations differ concerning what qualifies as the first complete derivation of  $E=mc^2$  and the date of its achievement—compare and contrast Refs. 71 and 72.
- <sup>70</sup> W. Rindler, *Introduction to Special Relativity*, 2nd ed. (Oxford U. P., Oxford, 1991), Secs. 26–27 (especially Sec. 27, wherein the two quotations cited from Sec. 6.3 of Ref. 69 are also given).
- <sup>71</sup> C. Lanczos, *The Einstein Decade (1905–1915)* (Academic Press, New York, 1974), pp. xi, 85–88, 96–105, 112, 139, 142–143, and 149–153. See especially pp. 85–86, 96–97, and 149–150.
- <sup>72</sup> H. C. Ohanian, *Einstein’s Mistakes* (W. W. Norton, New York, 2008), Chap. 7, and Notes and references for Chap. 7 on pp. 347–350.
- <sup>73</sup> H. Margenau, W. W. Watson, and C. G. Montgomery, *Physics: Principles and Applications* (McGraw-Hill, New York, 1949), Sec. 50.2.

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